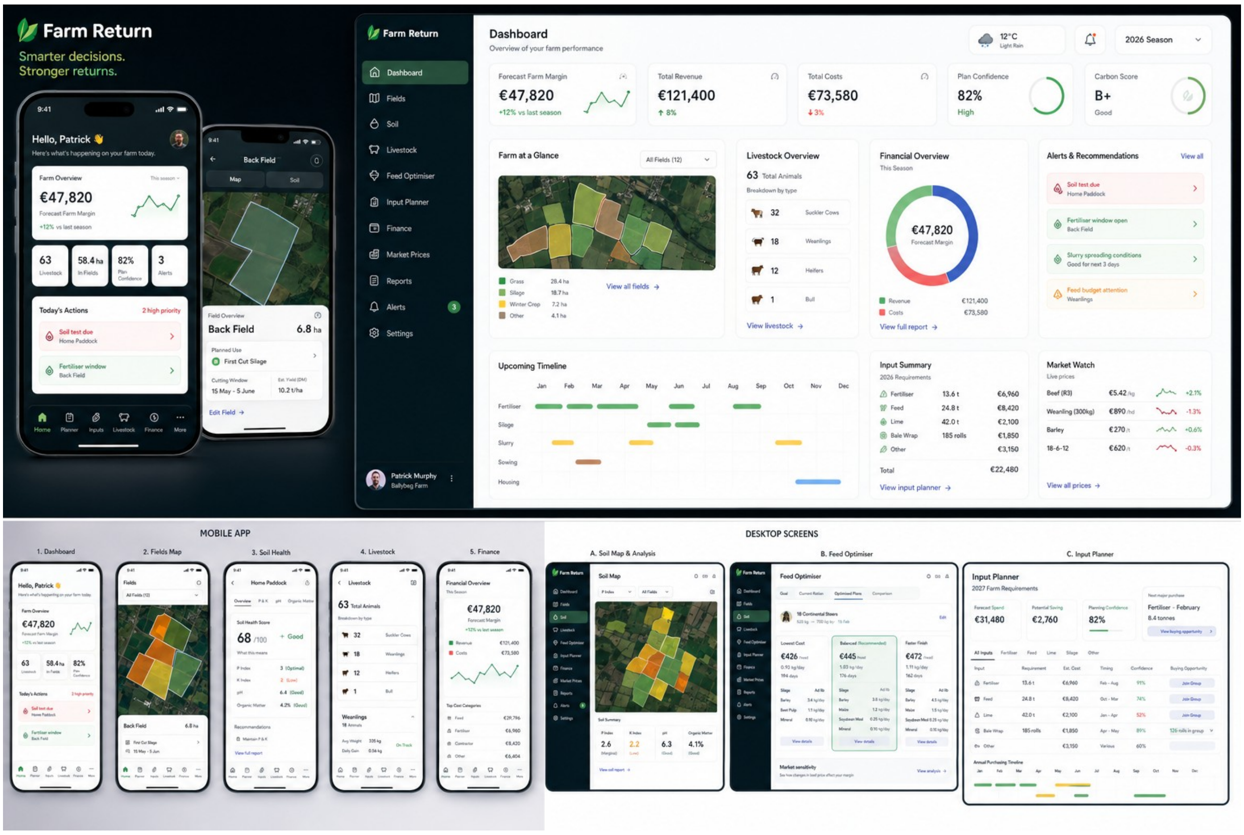
FARM RETURN

Product Build Specification & Visual Wireframe Pack

Claude Code source-of-truth document

Product mission

Build a free, premium-quality Irish farm management and financial intelligence platform that helps farmers understand their farm, forecast the inputs they will need, make better evidence-based decisions, and access lower input prices through aggregated bulk purchasing.



Approved visual direction - mobile and desktop product ecosystem. Sample names and numerical values are illustrative.

Version 1.0 | 23 August 2026 | Ireland

00 How to use this document

This document and the approved screenshots should be placed in the Farm Return repository before Claude Code begins implementation.

Source-of-truth hierarchy

1) This specification defines behaviour, data flow, architecture and build order. 2) The two combined product boards define the global visual language, navigation and responsive composition. 3) The individual detailed screen visuals define module-level content. If a generated number or sample name conflicts with the specification, the specification wins; if a layout conflict exists, the most detailed approved screen wins for that module.

All figures shown inside the wireframes are demonstration values. They are not production agronomic, nutritional, market or financial recommendations. Production calculations must be implemented from the evidence and rules framework described in this document.

Canonical product principles

- Enter once, use everywhere: never ask the farmer to re-enter data that Farm Return already holds or can derive.
- Automatic first, refinement second: use maps, public data and safe defaults to create an initial farm model, then let the farmer improve it progressively.
- Estimated, farmer-adjusted and verified data must remain visibly distinct and retain provenance.
- Science before AI: deterministic agronomic and nutritional engines produce the numbers; AI may explain, compare and summarise those outputs but must not invent them.
- Financial intelligence is free. The primary commercial model is bulk purchasing and transaction revenue, not paywalling core farm economics.
- Mobile and desktop are one product with a shared domain model and responsive UI, not separate applications.
- The approved UI should be reproduced closely; Claude Code must not redesign or simplify it without explicit instruction.

Contents

01 Product vision and commercial model

Farm Return is not a conventional subscription farm-recording application. The free product is intended to be genuinely valuable and to create the demand network that powers the purchasing business model.

Core proposition

Farm Return
A free farm-management, profitability and decision-support platform that uses mapped land, soil, livestock, forage, input and financial data to forecast farm requirements and aggregate farmer demand to reduce the cost of agricultural inputs.

Free product scope

Area	Free capability
Farm & fields	Farm profile; field mapping; areas; field names; land use; crop/silage intention; field history.
Soil & nutrients	Mapped soil profile; planning assumptions; farmer overrides; soil-test upload; fertiliser and lime requirements; cost estimates.
Livestock	Groups/categories; weights where known; housing assignment; breeding basics; current/forecast value; feed requirements.
Housing & slurry	Sheds; slatted storage; animal assignment; housing period; estimated slurry production; nutrient value and allocation.
Silage	Planned cuts; expected yield; actual harvest; pit/bales; silage analysis; winter forage balance; production cost.
Finance	Whole-farm revenue, costs, margin, cashflow, enterprise costs, livestock/feed/fertiliser cost views.
Weather/spreading	Field-level weather/soil-condition score, hard regulatory blockers, seasonal and near-term spreading windows.
Input Planner	12-month forecast of fertiliser, lime, feed, minerals, silage inputs and other requirements; timing, stock-on-hand and confidence.
Bulk purchasing	Forecast → farmer-confirmed → committed demand; regional groups; supplier tenders/offers; savings tracking.

Longer-term intelligence layer

- Advanced breeding, calving/lambing and replacement optimisation.
- Least-cost and profit-optimised feeding strategies using animal targets, forage analysis and current market prices.
- Sell-now versus finish/overwinter scenario modelling with sensitivity analysis.
- Multi-year whole-farm scenarios, benchmarks and AI-supported decision explanations.
- Additional integrations with laboratories, processors, marts, banks/accountants and farm sensors where commercially and legally appropriate.

02 Product architecture and connected farm model

The application must behave like one connected digital farm, not a collection of forms.

```
FARM
├─ Fields / Parcels
│  ├─ Soil profile + soil tests
│  ├─ Planned use: grazing / silage / crop
│  ├─ Nutrient requirement
│  ├─ Slurry / fertiliser applications
│  └─ Yield / harvest / field cost
├─ Livestock
│  ├─ Animal groups / individuals later
│  ├─ Weight / goal / market value
│  ├─ Housing assignment
│  └─ Feed requirement
```

```
├─ Housing
│   ├─ Shed type / capacity
│   ├─ Slurry storage
│   └─ Housing periods
├─ Forage & Feed Inventory
├─ Inputs & Stock on Hand
├─ Finance
└─ Plans / Forecasts
    ├─ Nutrient plan
    ├─ Feed plan
    ├─ Spreading plan
    └─ Input Planner -> Bulk Purchasing
```

Data precedence

Priority	Data status	Example
1	Verified	Laboratory soil test, silage analysis, measured weight, confirmed invoice/price.
2	Farmer adjusted	Farmer changes P/K assumption, drainage, price, yield or feed quantity.
3	Farm Return estimate	Calculated from farm data and evidence-backed assumptions.
4	Mapped/public source	Irish Soil Information System, Met Éireann data, public price benchmark.

Never overwrite provenance

When the farmer replaces an estimate with an actual value, retain the original value, source, timestamp and rule/model version. The working value changes, but history is not destroyed.

03 Data capture order and onboarding

The capture order is designed so each answer unlocks later calculations and reduces future input burden.

1. Create farm: farm name/location, primary enterprise(s), preferred units and only essential identifiers.
2. Map fields: draw/import boundaries; calculate hectares/acres automatically; assign simple field names.
3. Assign field use in bulk: grazing, first/second-cut silage, mixed, tillage/crop. Permit multi-select actions across fields.
4. Add livestock once: categories/groups and numbers; optional average weight/age/breed where known.
5. Add housing: choose shed type; link existing livestock groups to sheds; enter housing period; only request tank details if the farmer wants a refined estimate.
6. Review soil planning assumptions: mapped soil data is already populated; P/K assumptions can be batch edited; no repeated field-area entry.
7. Upload or enter soil tests: assign to mapped fields; verified values override planning assumptions.
8. Generate nutrient and organic-manure plan: use field use + soil + livestock/housing/slurry.
9. Generate silage/feed balance and financial forecast: silage intention automatically becomes expected forage production and feed value.
10. Generate Input Planner: all derived requirements are consolidated into forecast purchases; farmer confirms rather than retyping quantities.
11. Activate weather/spreading layer: field geometry, drainage and nutrient plan are already known; only live external conditions are added.

UX test

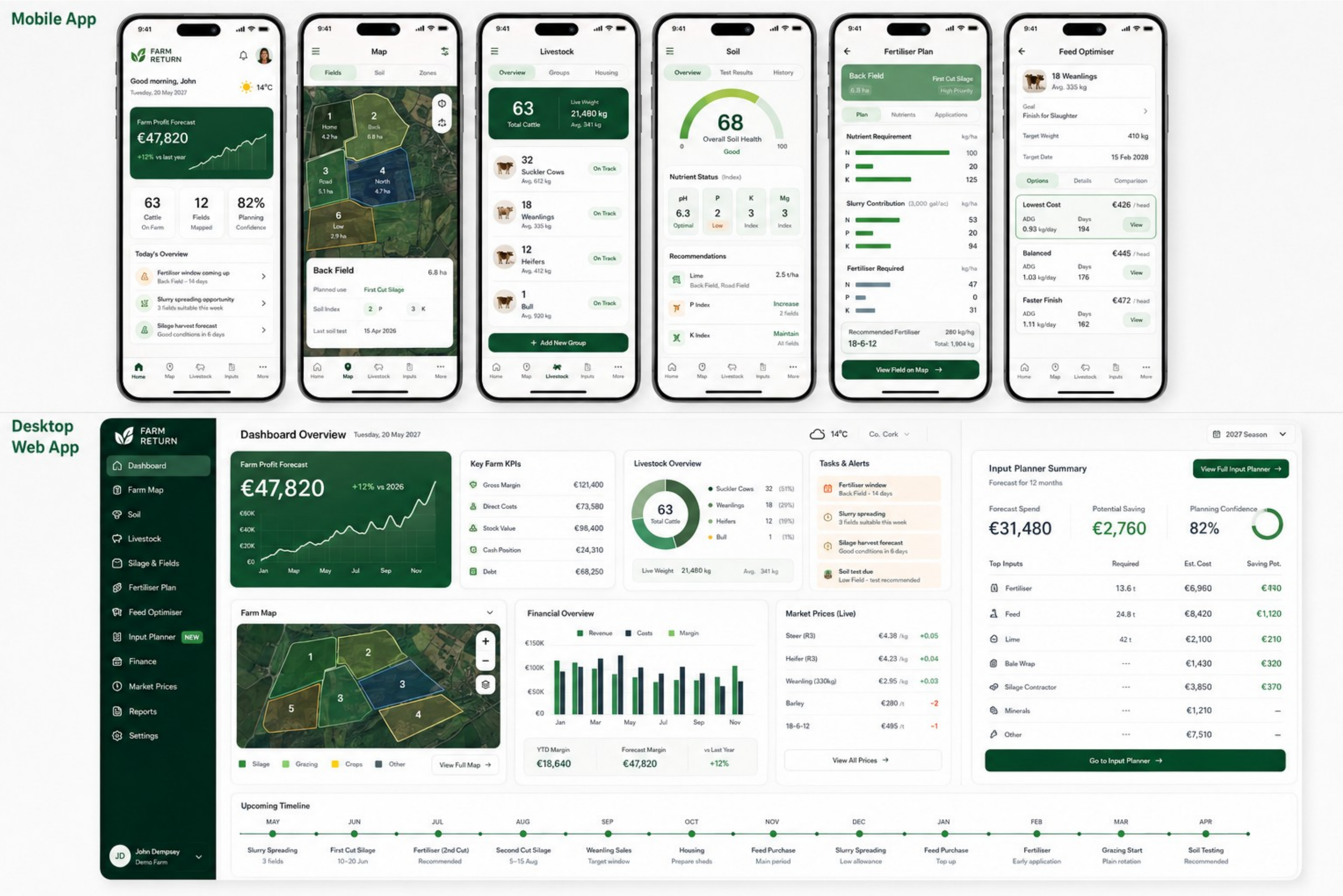
At every new form, Claude should ask: "Does Farm Return already know this? Can it derive it? Can it offer a safe assumption?" Only information that the farmer uniquely knows should be requested.

04 Screen inventory and navigation

The approved visual direction uses a map-led, finance-aware interface with clean white surfaces, dark green navigation and strong status colour semantics.

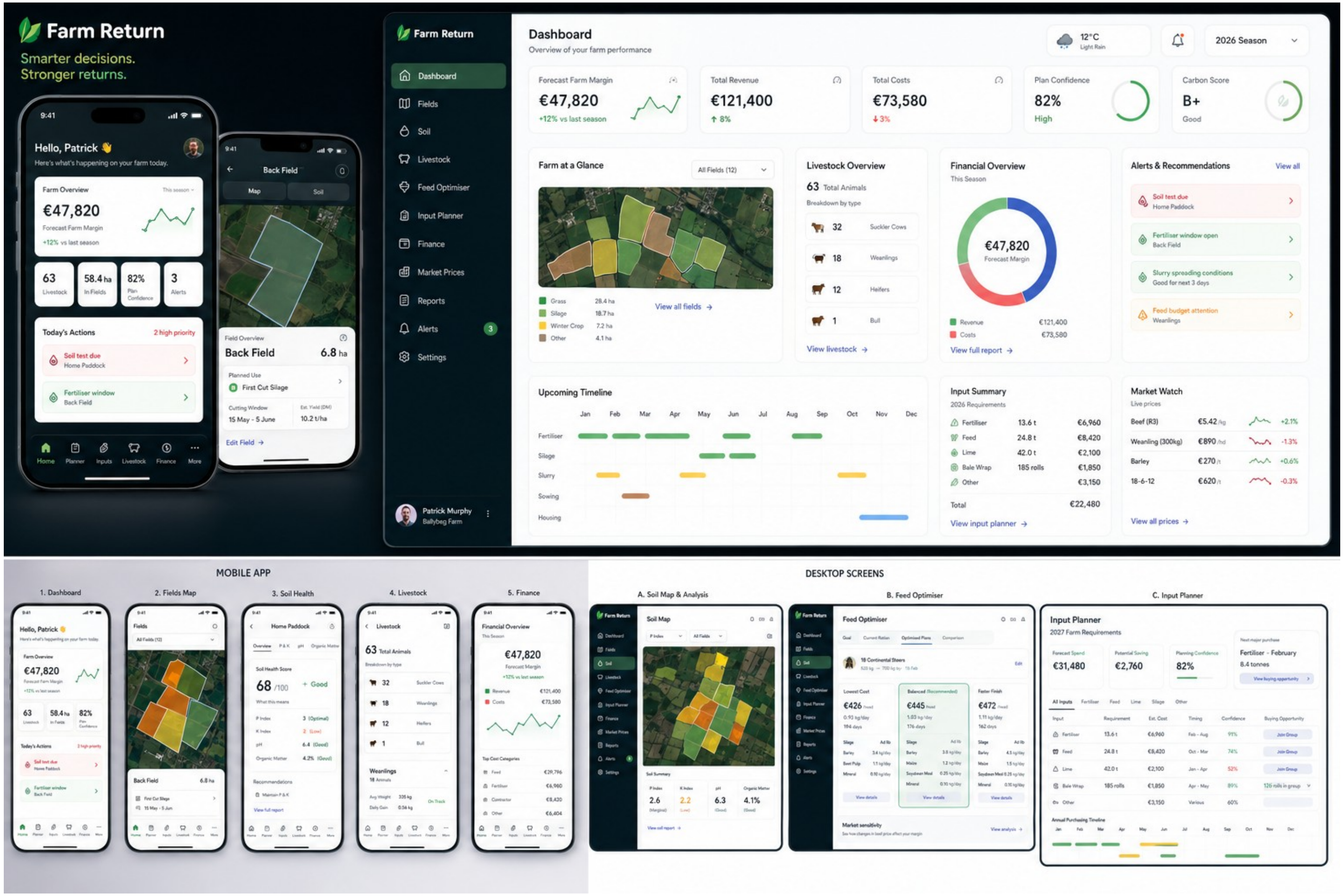
Route / Module	Mobile	Desktop	Primary purpose
/dashboard	Home	Dashboard	Farm-at-a-glance, financial forecast, actions, alerts, opportunities.
/fields	Map	Farm Map / Fields	Field boundaries, planned use, crop/silage allocation, per-field drilldown.
/soil	Soil	Soil	Mapped soil, P/K assumptions, verified tests, fertility status.
/livestock	Livestock	Livestock	Animal groups, numbers, weight/value, goals, housing link.
/housing	Contextual	Housing / Slurry	Shed assignment, slurry inventory, organic nutrient value.
/silage	Contextual	Silage & Fields	Cuts, expected/actual yield, forage balance and cost.
/nutrients	Nutrients	Fertiliser Plan	N/P/K requirement, slurry offset, products, field cost.
/spreading	Spreading	Spreading	0-100 field score, forecast windows, regulatory hard stops, application plan.
/feed-optimiser	Finance/context	Feed Optimiser	Least-cost / balanced / faster-finish strategies.
/input-planner	Inputs	Input Planner	12-month purchasing forecast and bulk-buy opportunities.
/finance	Finance	Finance	Revenue, cost, margin, cashflow, enterprise economics.
/market-prices	Context/More	Market Prices	Live/benchmark cattle, feed and fertiliser prices.
/reports	More	Reports	Farm plans, financial summaries, nutrient reports, exports.
/settings	More	Settings	Farm profile, units, data permissions, integrations, source preferences.

04A Approved overall desktop + mobile composition



Visual reference A - complete responsive product composition. Preserve navigation structure, map-led dashboard, Input Planner panel, timeline, price watch and card hierarchy.

04B Approved high-detail visual system



Visual reference B - premium visual system and responsive relationship between mobile and desktop. Sample labels and figures are illustrative.

05 Fields, soil and nutrient engine

The soil module must distinguish mapped physical soil properties from fertility assumptions and verified laboratory results.

Field soil record

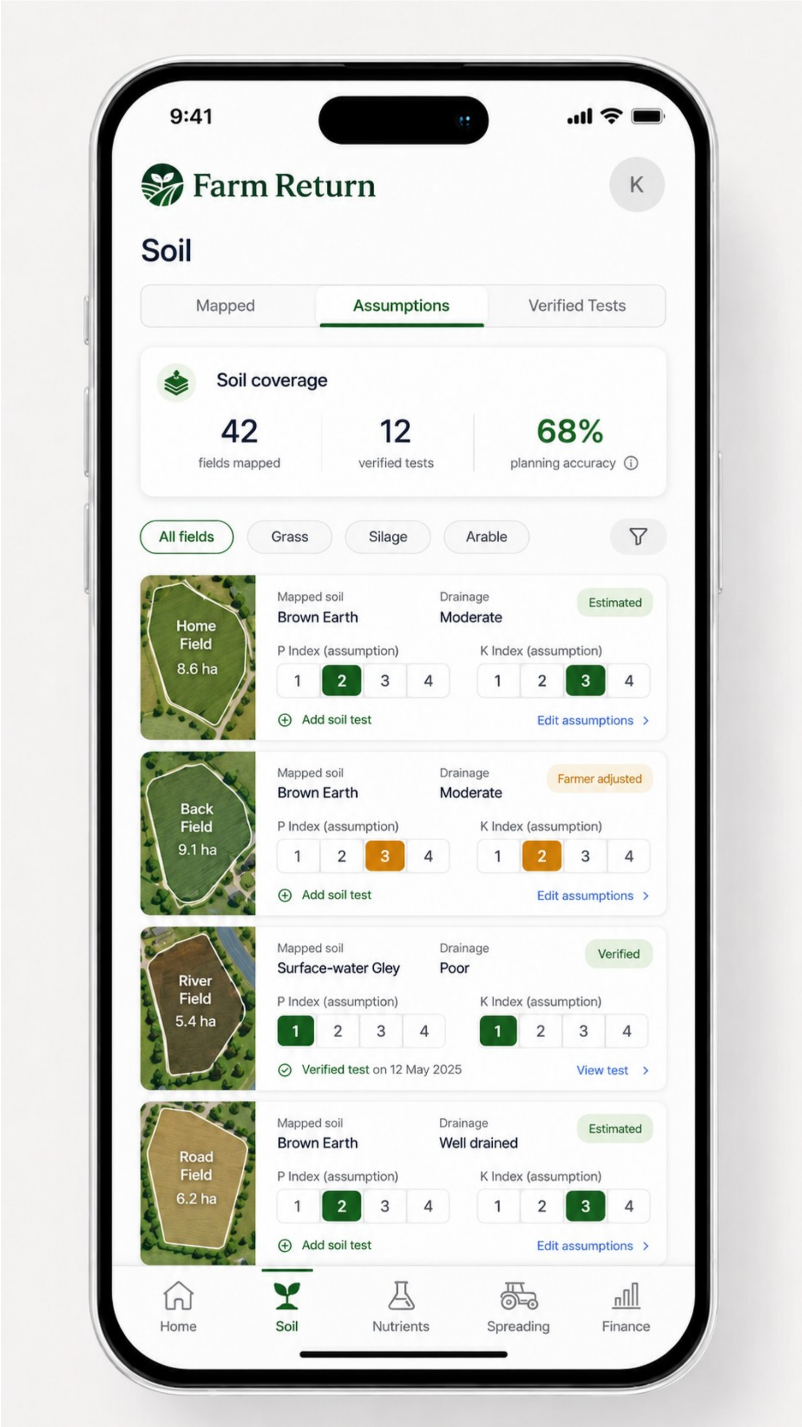
Layer	Captured/derived fields
Geometry	Field ID/name, polygon, area, centroid, optional LPIS reference, current/planned use.
Mapped soil	Soil association, dominant series/group, texture, drainage, depth, organic carbon/peat status, coverage percentage, dataset/version.
Planning fertility	P Index, K Index, optional pH planning value only if supportable, source="Farm Return assumption" or farmer-adjusted.
Verified test	Sample date, laboratory, field/sample reference, P, K, pH, lime requirement, optional Mg/organic matter and report file.
Calculation outputs	Gross N/P/K requirement, organic nutrient contribution, chemical top-up, recommended products, tonnes/kg and estimated cost.

Critical scientific rule

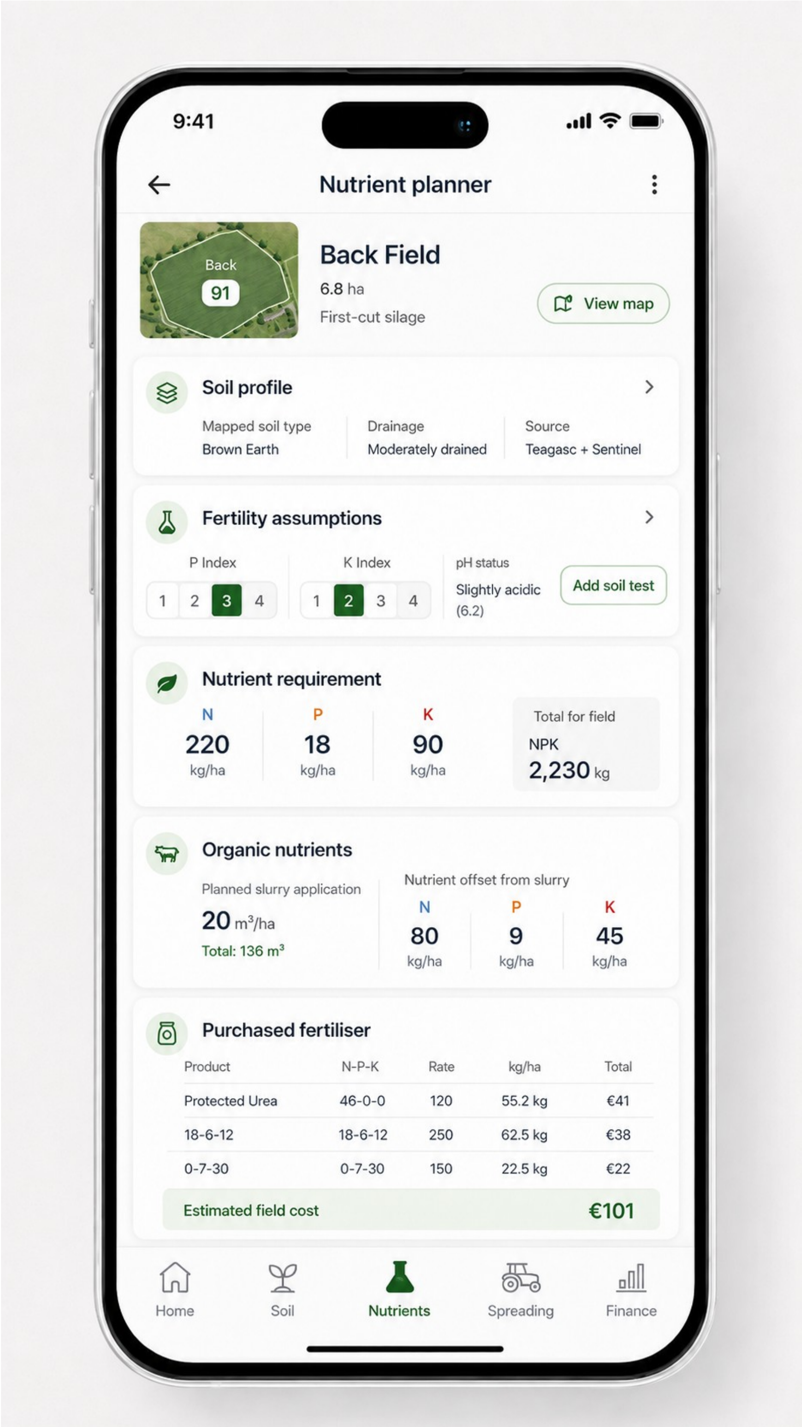
Do not infer an actual P or K Index from the national soil map. The map supplies physical soil context; P/K fertility is a planning assumption until adjusted or verified by soil analysis.

Calculation pipeline

```
Field geometry + planned use
         ↓
Mapped soil/drainage + P/K working values
         ↓
Crop/grass/silage nutrient requirement (versioned Irish rule set)
         ↓
Organic manure available and allocated
         ↓
Remaining N/P/K requirement
         ↓
Candidate fertiliser products / least-cost valid combination
         ↓
Quantity + timing + estimated € cost
         ↓
Input Planner forecast
```



Detailed mobile soil overview - mapped fields, assumptions, farmer-adjusted and verified states.



Detailed mobile nutrient planner - field context, soil, editable fertility assumptions, slurry offsets, purchased fertiliser and field cost.

06 Livestock, housing and slurry engine

Livestock numbers are entered once. Housing uses those existing groups; slatted housing creates an estimated slurry source that feeds nutrient and input planning.

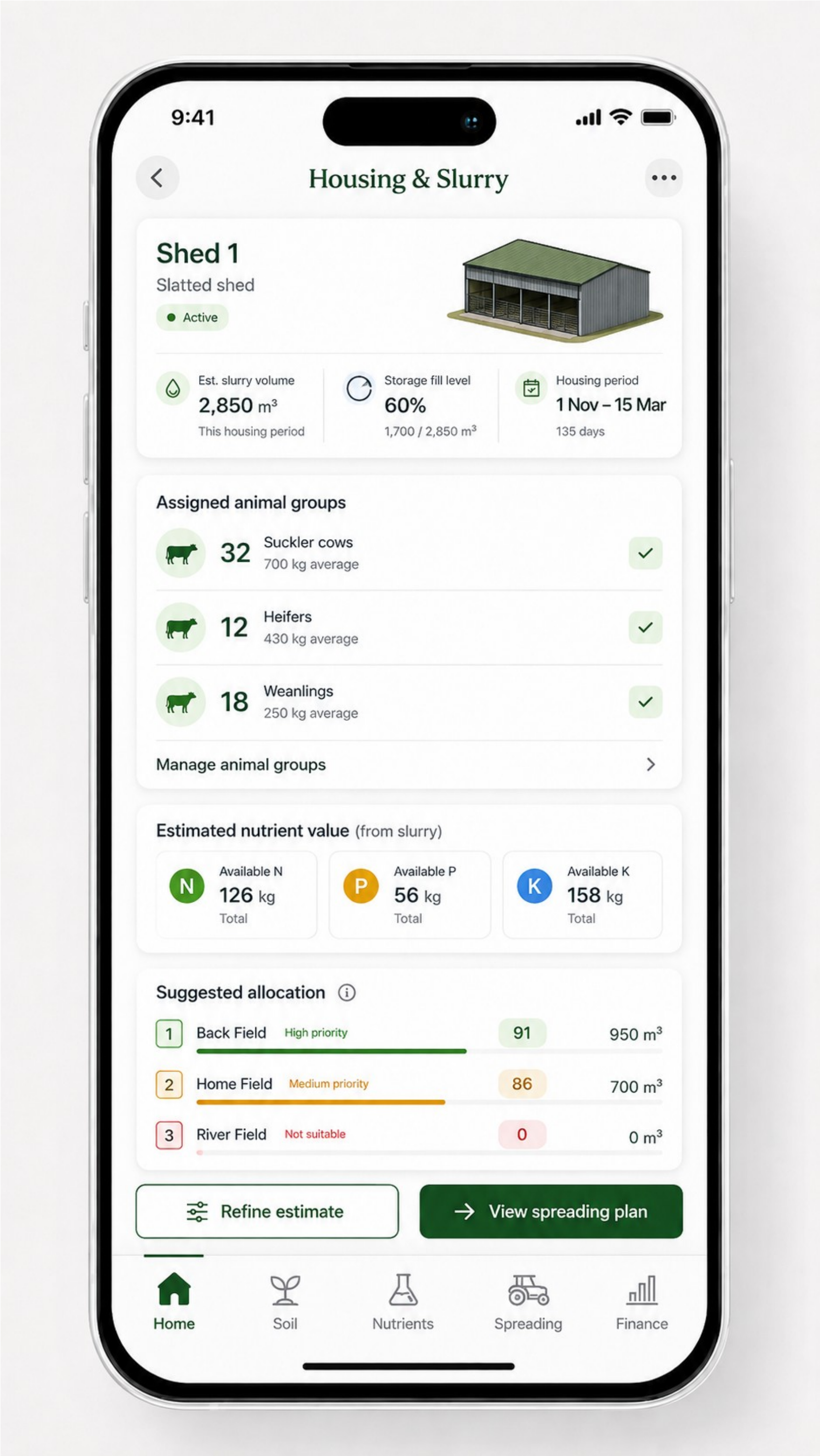
Livestock group MVP

Field	Purpose
Group/category	Suckler cows, dairy cows, bull, calves, weanlings, stores, steers, heifers; sheep equivalents later.
Count	Single source used across finance, housing, feed and forecasts.
Average weight / age	Optional initially; valuable for feed-cost and growth models.
Breed/type / sex	Optional or group-level; supports advanced feed/finishing

Field	Purpose
	logic.
Current system	Grazing/housed and housing assignment.
Goal	Maintain, grow, breed, sell as store, finish for slaughter; advanced mode.
Value	Public-market estimate, farmer adjustment, actual sale/purchase later.

Slatted shed logic

- When a farmer selects slatted shed, link the shed to previously entered animal groups rather than asking for animal numbers again.
- Estimate slurry production from animal category and housing duration using the current regulatory/storage coefficient set; mark it as estimated.
- Allow optional refinement using tank dimensions/capacity, observed fill level, dilution/soiled water and slurry analysis.
- Convert estimated/analysed slurry to available N/P/K using a versioned organic-manure rule set, with application method and timing considered.
- Allocate slurry to fields according to nutrient need and agronomic/regulatory constraints before calculating chemical fertiliser top-up.



Housing & slurry screen - linked livestock groups, estimated storage, nutrient value and suggested field allocation.

07 Silage and winter-feed engine

Silage is a field-level production decision that must automatically affect nutrient demand, feed availability and farm economics.

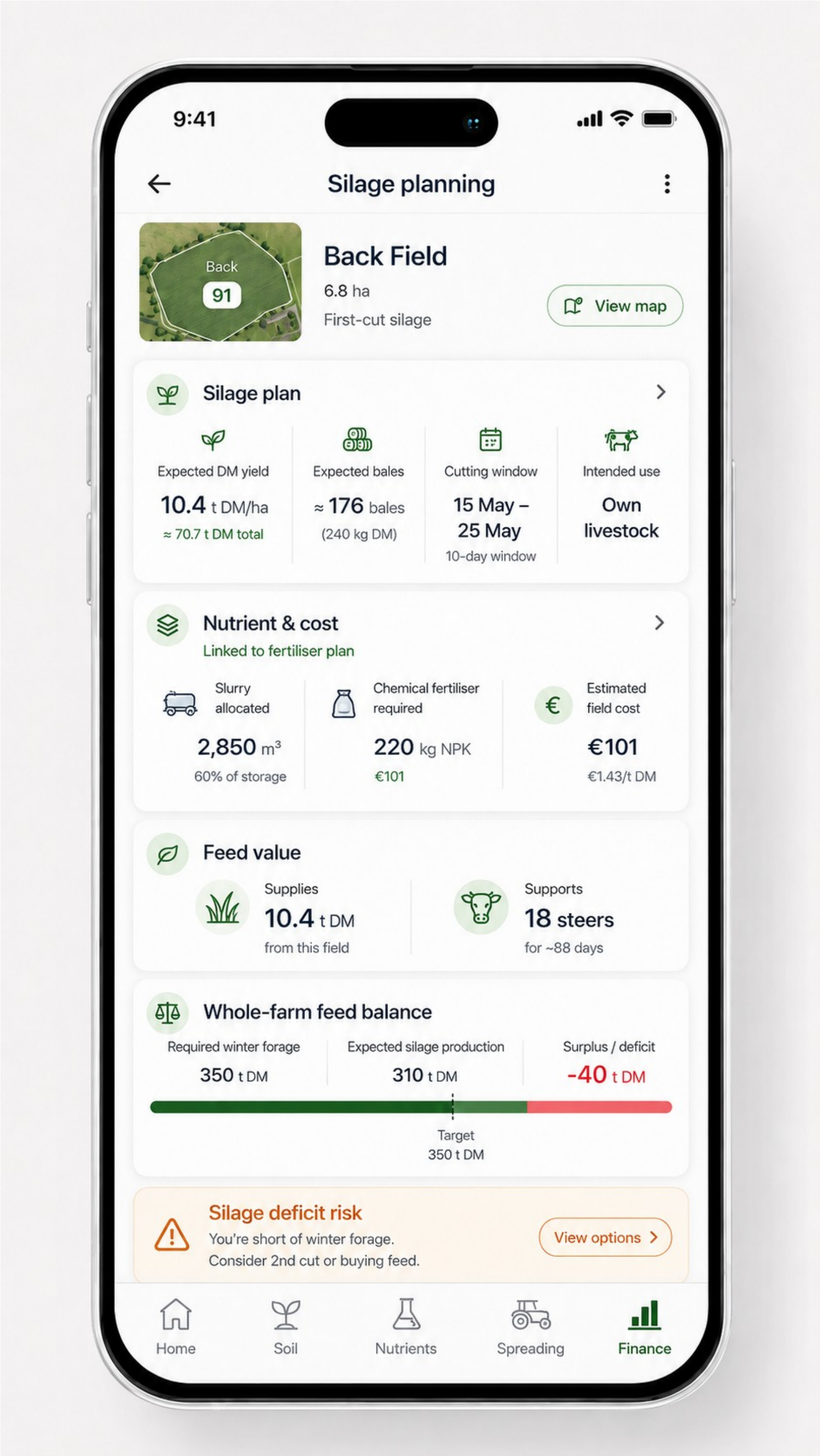
Silage data entered once

Data	Capture behaviour
Planned cut	First, second, third; farmer choice; bulk edit across mapped fields.
Harvest system	Pit or bale.
Target cut window	Farm Return suggestion based on system/science; editable by farmer.
Expected yield	Evidence-backed estimate; farmer can adjust; replaced

Data	Capture behaviour
	by actual harvest.
Expected quality	Planning assumption such as DMD; upload lab analysis for verified value.
Intended use	Own livestock, sale or both.
Actual output	Tonnes/pit estimate or bale count; moisture/DM if known.
Production cost	Fertiliser/slurry, contractor/machinery, wrap/bales and other direct costs.
Feed inventory	Automatically convert output to usable forage inventory and feed cost basis.

Cross-module effects

- Marking a field for silage changes the relevant nutrient recommendation and raises the priority for recycling slurry P and K where appropriate.
- Expected silage dry matter becomes an asset in the winter-feed budget, reducing predicted purchased feed if sufficient.
- Actual silage analysis changes feed-optimiser concentrate requirements and therefore livestock input cost and projected margin.
- If winter feed is forecast to be short, Farm Return can compare second-cut production, purchased forage and concentrate-replacement scenarios.
- Silage inputs such as wrap and contractor requirements can appear in Input Planner when the relevant harvest system is selected.



Silage planning - field plan, expected production, nutrient/cost link, feed value and whole-farm deficit warning.

08 Financial intelligence

The financial side is a free core capability and must be fed automatically by the operational farm model.

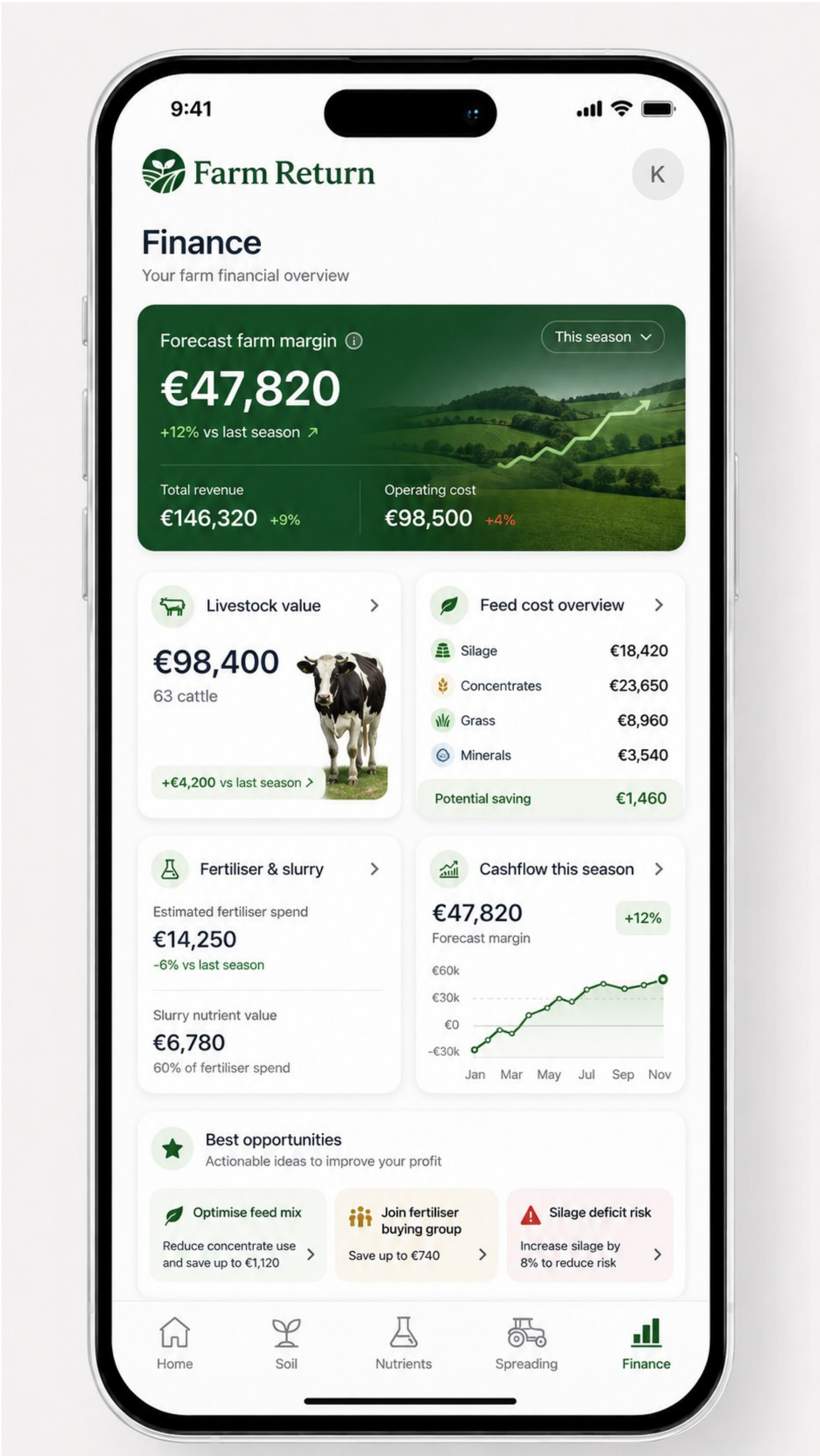
Whole-farm financial model

Area	Examples of calculated / captured values
Revenue	Livestock sales/value change, crop/silage sales, subsidies/other farm income, actual transactions later.
Feed	Silage production cost, concentrates, straights, minerals, purchased forage, grazing cost assumptions.
Fertiliser & lime	Nutrient-plan product quantities x benchmark/farmer/bulk-buy prices; slurry nutrient value

Area	Examples of calculated / captured values
	shown separately.
Livestock	Estimated/current value, feed cost/head/day, cost to target, sale scenario margin.
Field/enterprise	Cost/ha, feed production cost/t DM, gross margin, direct cost by field/enterprise.
Cashflow	Expected timing of major input purchases and expected sales/income.
Opportunity layer	Potential input savings, alternative feed strategies, bulk-buy savings, feed deficit actions.

Price provenance

Each price must have a source and date: public benchmark, farmer price, invoice/contract, supplier quote or Farm Return bulk price. Calculations must automatically refresh when the active price source changes.



09 Feed optimiser and livestock economics

The advanced optimiser should solve for a target outcome under nutrition, intake, animal-performance and market constraints; it should not be a generic AI ration suggestion.

Free livestock cost layer

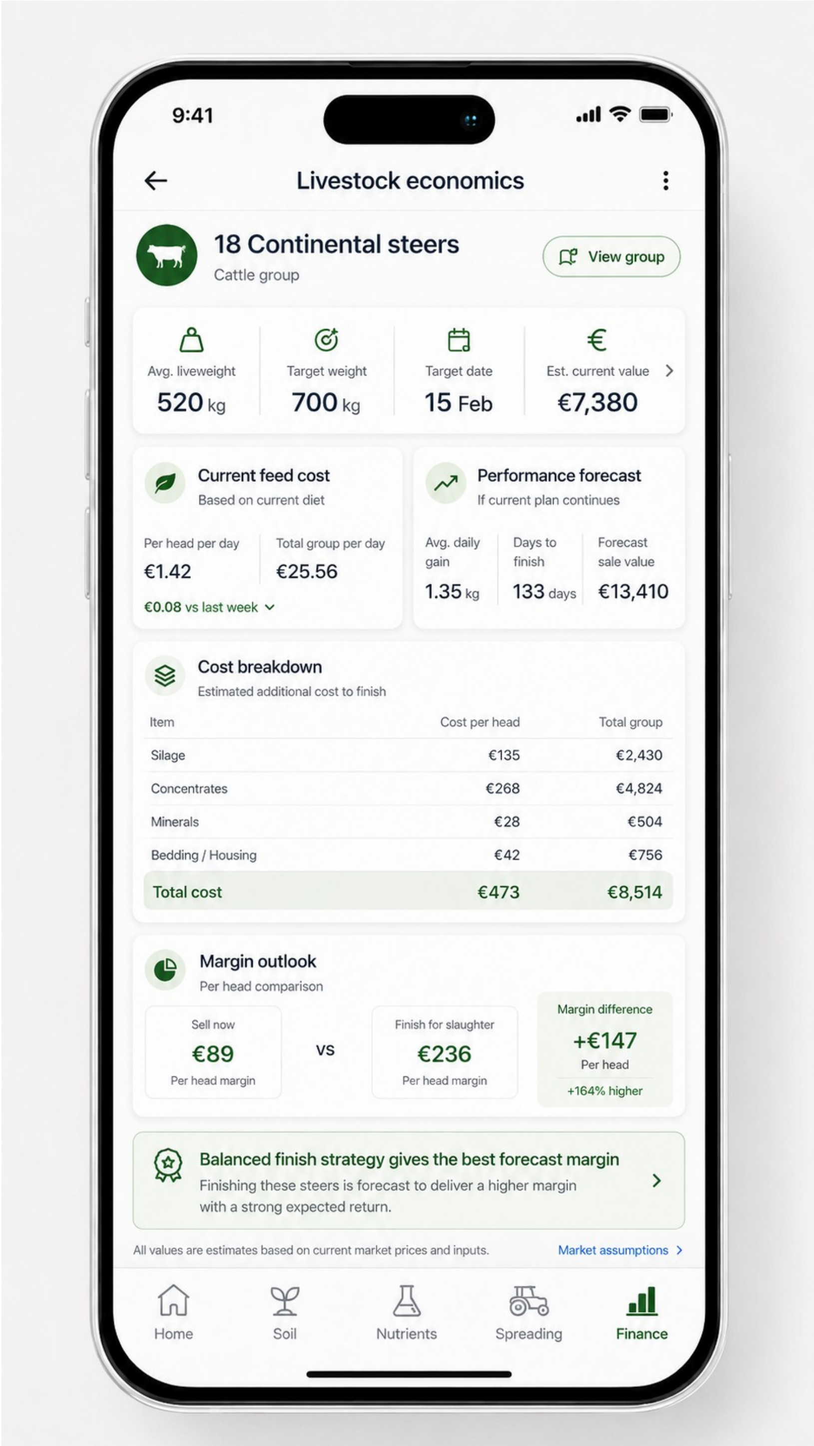
- Use livestock groups, housing period, forage inventory/quality and benchmark feed prices to estimate feed cost by group, head/day and season.
- Allow the farmer to replace public benchmark prices with their actual merchant/contract price once; reuse it everywhere.
- Use actual silage analysis when available; otherwise use a clearly labelled planning assumption.
- Financial screen should show the cost driver, not only total spend: silage, concentrate, grass, minerals, bedding/housing and other variable costs.

Advanced optimiser

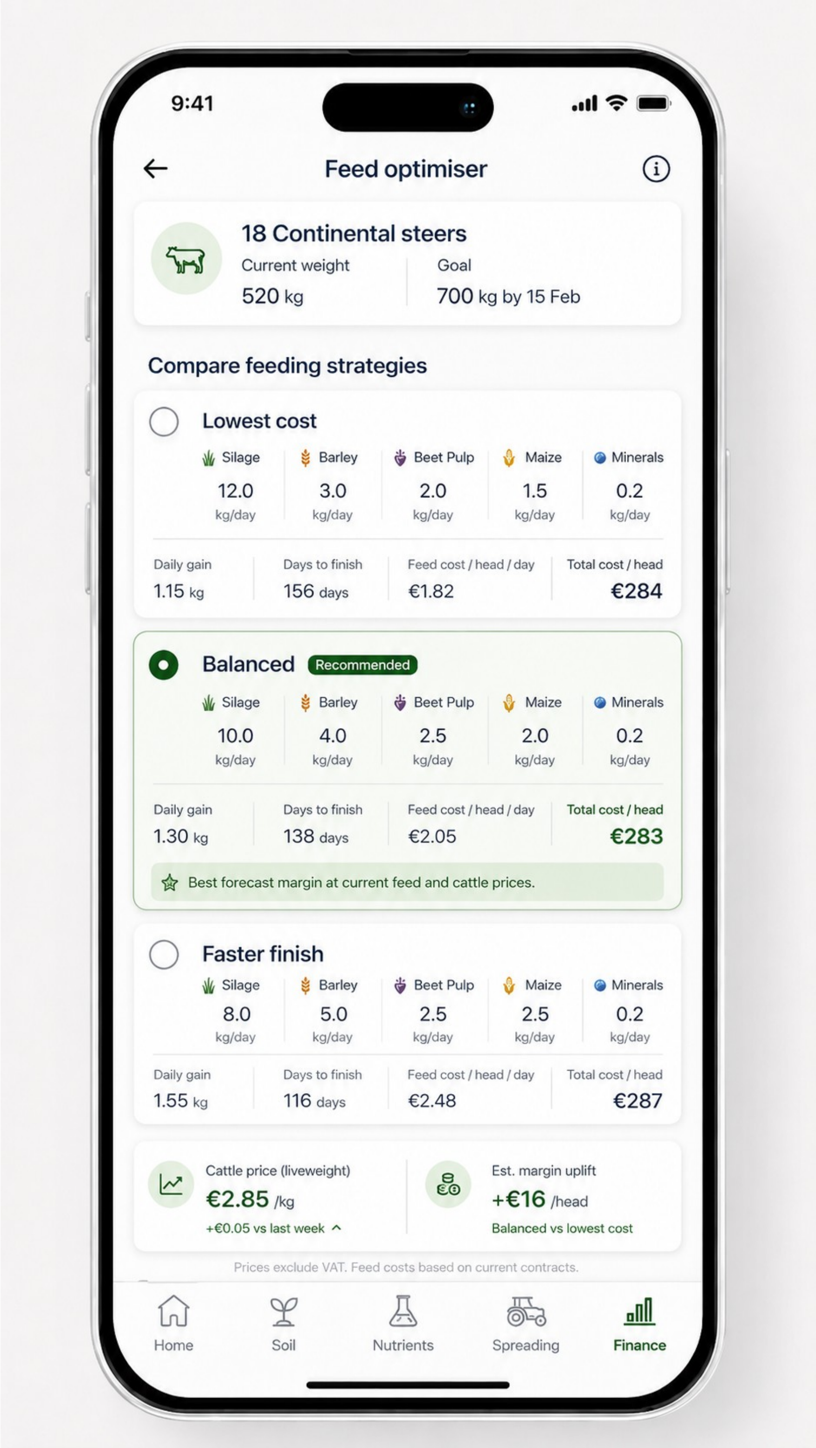
Input	Examples
Animal state	Group, sex/type, age, current weight, body condition/fat score where available, breed category.
Goal	Target weight/carcass, target date, sell as store, finish for slaughter, target daily gain.
Forage	Silage/grass inventory, DM/DMD/energy/protein analysis, cost basis.
Purchased feeds	Barley, maize, beet pulp, soya/other protein, compounds, minerals; nutrient profile and current price.
Constraints	Expected DMI, minimum fibre/forage, protein/energy requirements, safe inclusion ranges, mineral needs.
Market	Current cattle price by relevant grade/category; sensitivity range and price date.
Objective	Lowest feed cost, balanced/highest forecast margin, or faster finish.
Output	Ration options, kg/head/day, expected ADG, days to finish, feed cost/head, expected sale value, margin and sensitivity.

Optimisation principle

Optimise profit, not just price per tonne. A slightly more expensive daily ration can produce a better margin if it reaches market specification sooner or uses shed capacity more efficiently.



Livestock economics - current weight/value, feed cost, performance forecast, cost to finish and sell-now versus finish margin.



Feed optimiser - lowest cost, balanced and faster-finish strategies with performance and cost comparisons.

10 Spreading conditions engine

Every mapped field can receive a live suitability score, but regulatory and environmental blockers must override any numerical score.

Two related scores

- Slurry/organic manure spreading score.
- Chemical fertiliser spreading score.
- Both use the same field geometry, soil/drainage, planned application, crop demand and live weather inputs, but can use different thresholds/weights.

Hard-stop rules

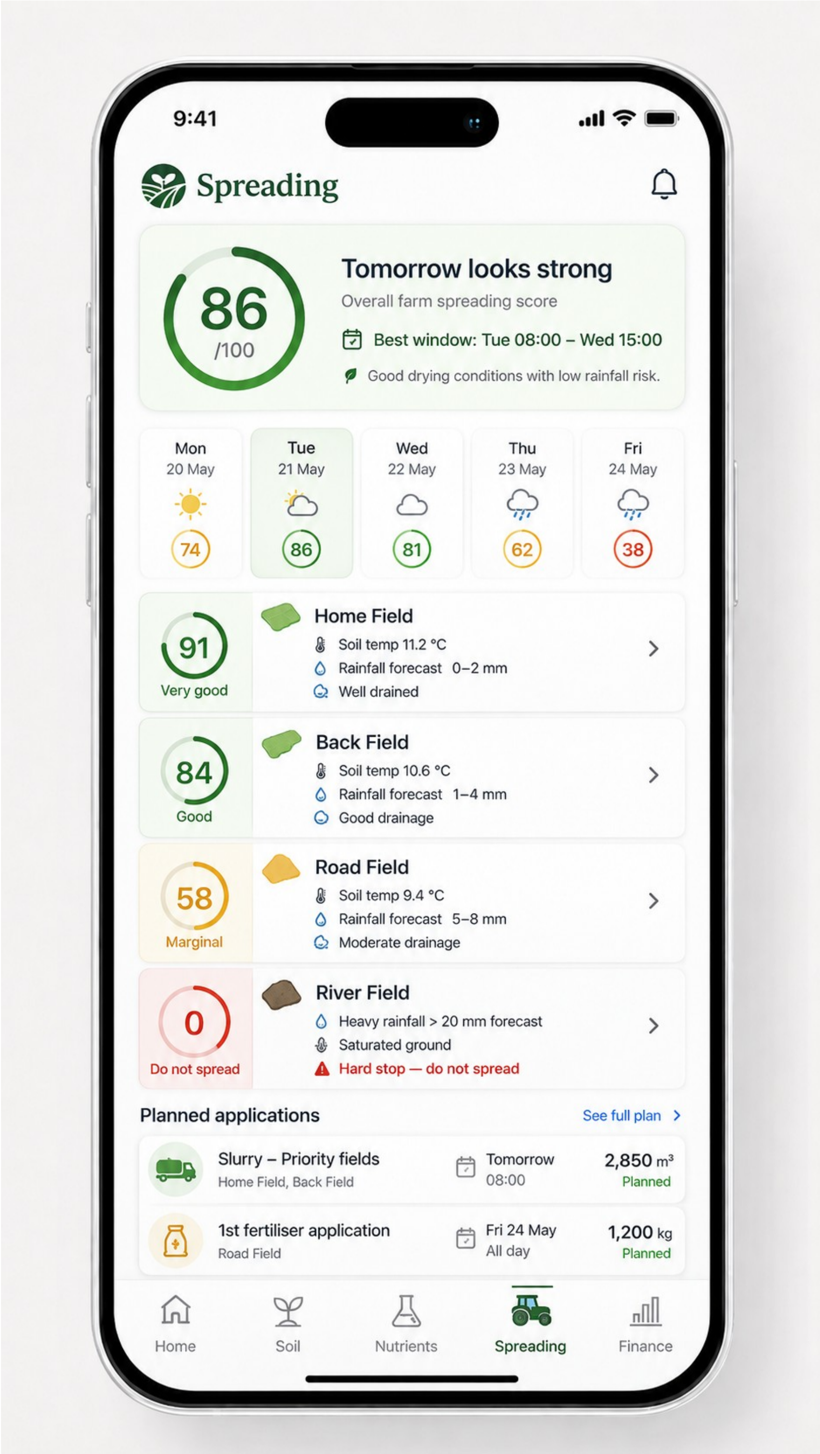
Where current Irish rules prohibit or make spreading unsafe (for example closed periods or specified wet/flooded/frozen/heavy-rain conditions), the UI must show a hard stop such as "0 - Do not spread" with the reason. A hard stop is not merely a low score.

Indicative score components - validate before production

Component	Role
Forecast rainfall	Near-term wash-off/runoff risk; regulatory forecast requirement where applicable.
Soil moisture / trafficability	Use drainage class and available SMD/modelled wetness; prevent damage/runoff.
Soil temperature & trend	Agronomic uptake context; never present a modelled station value as an in-field sensor measurement.
Crop/grass demand	Nutrient uptake window and planned application need.
Drainage/topographic/environmental risk	Field-specific risk modifiers and required buffers.
Application conditions	Wind/operational suitability and product/application method where relevant.

Time horizons

- Seasonal plan: target weeks/periods and expected quantity based on nutrient need.
- Near-term planner: 5-10 day field score forecast.
- Live action: best current window; when application is marked complete, update remaining nutrient requirement and input stock.



Spreading screen - farm score, multi-day outlook, field-level reasons, hard stop and planned applications.

11 Input Planner and bulk purchasing

This is the commercial heart of Farm Return: turn the connected farm model into forecast demand, then aggregate that demand to win better prices.

Input Planner inputs

Farm data	Forecast produced
Mapped fields + soil + planned use	Fertiliser, lime and crop/silage input volumes.
Livestock + housing	Feed/mineral demand, bedding where relevant, slurry production.
Silage plan + actual forage inventory	Purchased feed requirement after own-feed contribution.
Feed optimiser	Potential straights/compound requirements under selected livestock strategy.
Existing stock on hand	Subtract stock already owned from purchasing

Farm data	Forecast produced
	requirement.
Historical/benchmark/current prices	Forecast cost and savings range.
Calendar/season	Required-by month/window for procurement aggregation.

Demand states

State	Meaning	Commercial use
Forecast	Farm Return estimates the farmer will need the quantity.	Network demand signal; not a commitment.
Farmer confirmed	Farmer reviews and confirms/adjusts the forecast.	Higher-confidence demand pool.
Committed to buying group	Farmer opts into a specific procurement event/offer.	Supplier tender/negotiation quantity.
Purchased	Accepted quote/order recorded.	Revenue, savings and stock update.

Required page components

- Forecast spend, potential savings and planning-confidence headline cards.
- Category breakdown: fertiliser, feed, lime, minerals, silage inputs, contractor/other where meaningful.
- Quantity, estimated cost, timing and confidence for each line.
- Annual purchasing timeline showing concentration of demand by month.
- Stock-on-hand deduction: required quantity - existing stock = purchase requirement.
- Bulk-buy opportunity cards showing user requirement, regional confirmed/committed demand, target/current price and potential saving.
- One-tap quantity confirmation and adjustment instead of recreating an order from scratch.
- Post-purchase savings ledger and input-stock update.

Network effect

The product loop is: better farm data → better demand forecast → larger buying groups → stronger supplier pricing → measurable farmer savings → greater retention and more complete data.

12 Design system and responsive behaviour

The implementation should feel like high-end fintech/consumer software applied to farming, not a generic admin dashboard.

Visual language

Element	Canonical direction
Typography	Inter / Inter Display family; bold clean numeric hierarchy; compact secondary labels.
Primary colour	Deep forest green navigation and key financial/actions; green used purposefully rather than washing the entire UI.
Surfaces	White/off-white background, subtle neutral borders, low-elevation soft cards, generous whitespace.
Status colour	Green = good/optimal/confirmed; amber = attention/marginal; red = blocked/risk; blue = information/data/weather.
Corners	Rounded cards and controls; refined rather than playful.
Map	Satellite/aerial field map as a hero surface; crisp field outlines; selectable persistent field state.
Charts	Simple, calm, legible; avoid decorative 3D effects; status/financial insight first.
Icons	Consistent line icon set such as Lucide; agricultural-specific assets only where they improve recognition.
Motion	Subtle transitions and drawers; respect reduced-motion; no animation that hides information.
Density	Mobile is focused and vertically progressive; desktop uses multi-column overview without turning into a spreadsheet wall.

Mobile navigation

Canonical bottom navigation: Home, Map/Fields, Livestock, Inputs/Planner and contextual/More depending on the final shell. The detailed module pages may expose Soil, Nutrients, Spreading and Finance as direct tabs or contextual routes. Preserve the visual hierarchy shown in the approved boards and do not create a second competing navigation pattern.

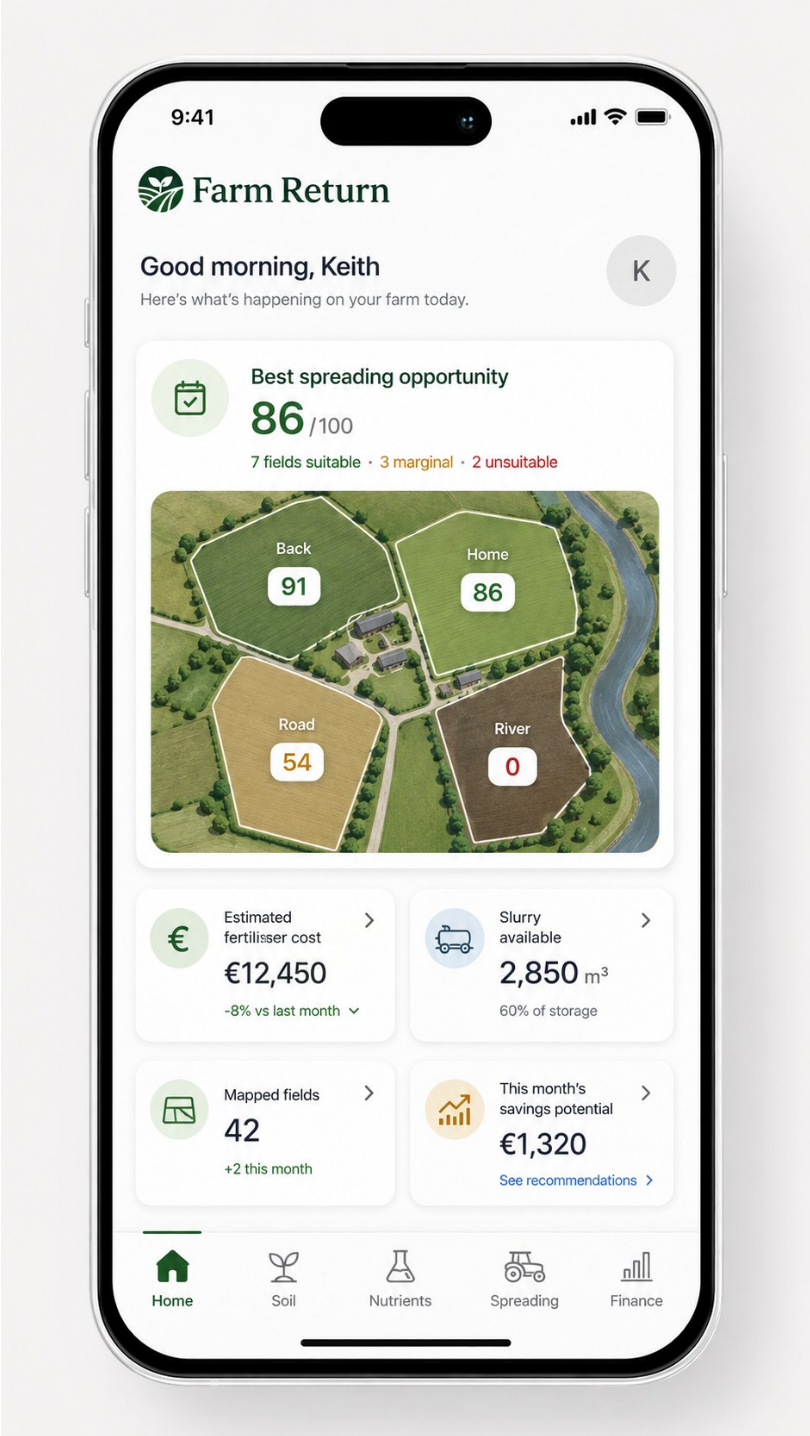
Desktop navigation

Dark-green left rail: Dashboard, Fields/Farm Map, Soil, Livestock, Silage & Fields, Fertiliser Plan/Nutrients, Feed Optimiser, Input Planner, Finance, Market Prices, Reports and Settings. Spreading can be first-class if the final route inventory includes it separately.

Responsive rule

One product, two compositions

Do not merely scale the mobile layout wider. Desktop should adopt the approved multi-column dashboard, persistent left navigation and side-by-side analytical cards, while sharing the same components, design tokens and data model.



Detailed mobile dashboard reference - spreading opportunity, map hero and farm-level KPI cards.

13 Technical architecture and repository structure

Start with a UI-first MVP using realistic mock data; domain calculations and integrations are added after the approved visual shell is stable.

Recommended stack

Layer	Recommendation
Application	Next.js + TypeScript.
Styling	Tailwind CSS with Farm Return tokens; shadcn/ui only as low-level primitives, not default styling.
Icons	Lucide; custom agricultural icons only when needed.
Charts	Recharts or equivalent deterministic chart library.
Maps	MapLibre/Mapbox or suitable satellite mapping provider; field polygon drawing/editing.

Layer	Recommendation
Forms	React Hook Form + Zod.
Database/auth	Supabase/Postgres once the mock-data UI is approved.
Domain engine	Pure TypeScript modules/functions with versioned rules and unit tests; no calculations embedded in React components.
Testing	Vitest/unit tests for domain logic + Playwright E2E and visual regression.
Deployment	Vercel initially; environment variables for API/provider keys.

Repository shape

```
farm-return/
├─ design/
│  ├─ reference/mobile/
│  ├─ reference/desktop/
│  ├─ design-system.md
│  └─ screen-specification.md
├─ docs/
│  ├─ product-requirements.md
│  ├─ data-model.md
│  ├─ agronomy-engine.md
│  ├─ feed-engine.md
│  ├─ finance-engine.md
│  └─ evidence-register.md
├─ src/
│  ├─ app/{dashboard,fields,soil,livestock,housing,silage,nutrients,spreading,feed-optimiser,input-
planner,finance,market-prices,reports,settings}/
│  ├─ components/{ui,farm,maps,charts,finance}/
│  ├─ domain/{farm,soil,nutrients,livestock,slurry,silage,feed,finance,inputs}/
│  ├─ data/mock-farm.ts
│  └─ lib/
├─ tests/{unit,e2e,visual}/
├─ CLAUDE.md
└─ package.json
```

Core reusable components

- AppShell, DesktopSidebar, MobileBottomNav, PageHeader.
- MetricCard, FinancialHeroCard, OpportunityCard, AlertCard, StatusBadge, ConfidenceBadge, SourceBadge.
- FarmMap, FieldPolygonLayer, FieldCard, FieldDrawer, MapLegend.
- SoilIndexSelector, SoilSourcePanel, TestUploadCard.
- NutrientBars, OrganicNutrientCard, ProductRecommendationTable.
- LivestockGroupCard, GoalCard, EconomicsSummary.
- ScoreRing, ForecastDayCard, HardStopAlert.
- InputRequirementRow, PurchaseTimeline, BuyingOpportunityCard.
- MarketPriceRow, CashflowChart, MarginComparison, SensitivityTable.

14 Claude Code implementation workflow

Claude Code should be used as the primary builder, but it must operate against this specification and the visual references rather than improvising product decisions.

Step 1 - initialise in plan mode

```
claude --permission-mode plan
```

Step 2 - first repository prompt

```
We are building Farm Return, a premium-quality Irish farm management,
financial intelligence and input-planning platform.

DO NOT WRITE APPLICATION CODE YET.

Read the full /docs product specification and every image in /design/reference.
The approved references are the visual source of truth. Do not redesign,
simplify or substitute a generic SaaS dashboard.
```

```
Core rule: the farmer enters information once and Farm Return reuses it
everywhere. Derived/public data should be populated automatically where safe.
Estimated, farmer-adjusted and verified values must remain distinct.

For this task:
1. Audit the planned route map and responsive navigation.
2. List every reusable component visible in the references.
3. Convert the approved visual language into design tokens.
4. Identify any conflict between screenshots and this specification.
5. Propose the Next.js folder/component architecture.
6. Propose the mock-data model for the UI phase.
7. Propose Playwright screenshot tests at mobile and desktop viewports.
8. Produce the screen-by-screen implementation order.

Do not build until I approve the plan.
```

Step 3 - create CLAUDE.md rules

- Never remove an approved screen element or feature without explicit instruction.
- Never alter the Farm Return design system to a stock framework look.
- Reuse existing components; do not create visually similar duplicates.
- Every screen is reviewed at mobile and desktop sizes.
- Do not place agronomy, feed or financial formulas in React components.
- No production scientific number may be invented by the model; implement only documented rules with tests and source/version metadata.
- Never ask for data already available elsewhere in the central farm model.
- Before a new field is added to a form, identify where else it might already exist or whether it can be derived.
- All external data must retain provider, timestamp, licence/terms considerations and fallback behaviour.

Step 4 - build UI screen-by-screen

12. Desktop Dashboard + Mobile Home.
13. Fields/Farm Map.
14. Soil.
15. Livestock.
16. Housing/Slurry.
17. Silage.
18. Nutrients/Fertiliser.
19. Spreading.
20. Finance.
21. Livestock Economics.
22. Feed Optimiser.
23. Input Planner.
24. Market Prices, Reports, Settings.

Step 5 - screenshot comparison prompt

```
Run the application and capture this screen at the approved mobile and
desktop viewport sizes. Compare the render with the matching image in
/design/reference.

Do not add or remove features. Identify discrepancies in layout, spacing,
typography, colour, radius, card dimensions, map sizing, icon scale,
hierarchy and responsive behaviour. Correct them and repeat until the
implementation is materially consistent with the approved visual reference.

Do not accept a generic approximation.
```

Step 6 - only then implement real engines

Replace mock outputs one domain at a time: soil/nutrients → livestock/housing/slurry → silage/feed inventory → finance → spreading → feed optimiser → Input Planner/bulk buying. Each engine receives deterministic tests and an evidence/version record before its values are allowed onto production screens.

15 Quality, science and regulatory governance

Farm Return will be trusted only if a farmer can understand what a number means, where it came from and how current it is.

Every material recommendation should carry metadata

Metadata	Example
Value	P Index working value = 2.
Status	Farmer adjusted / verified / estimated.
Source	Soil test; Teagasc rule; Irish Soil Information System; Met Éireann; CSO; supplier quote.
Source date/version	Dataset or rule publication/version and retrieval date.
Calculation version	nutrient_engine_v1.2.0.
Confidence	High/medium/low where estimate uncertainty is meaningful.
Regulatory status	Planning advice vs compliance value/rule.
Explanation	Plain-English “Why?” panel without overstating certainty.

Scientific governance rules

- Use Teagasc nutrient and livestock research as the primary Irish agronomic/nutritional evidence base, supplemented by current statutory requirements.
- Treat the 1:250,000 Irish soil map as a broad physical soil context, not a substitute for field soil testing.
- Keep agronomic recommendations separate from statutory compliance logic; the rules engine should be versioned and updateable.
- For weather/spreading, use authoritative Irish weather/agrometeorological data and never claim station/model data is an in-field sensor measurement.
- Public prices are benchmarks. Display date and source, and allow farmer actual prices or supplier/bulk-buy quotes to supersede them.
- Have an Irish agronomist/nutritionist review production formulas and scoring thresholds before launch; automated tests should lock verified examples.

Visual QA acceptance criteria

- All approved UI elements present and navigable.
- No generic default component styling where a Farm Return reference exists.
- No text clipping or overflow at target mobile/desktop sizes.
- Keyboard/focus accessibility and touch targets appropriate for mobile.
- Core flows usable without hover-only interactions.
- Visual regression screenshots checked before merging major screen changes.
- Mock demo values clearly separated from real engine output during development.

16 Delivery phases and roadmap

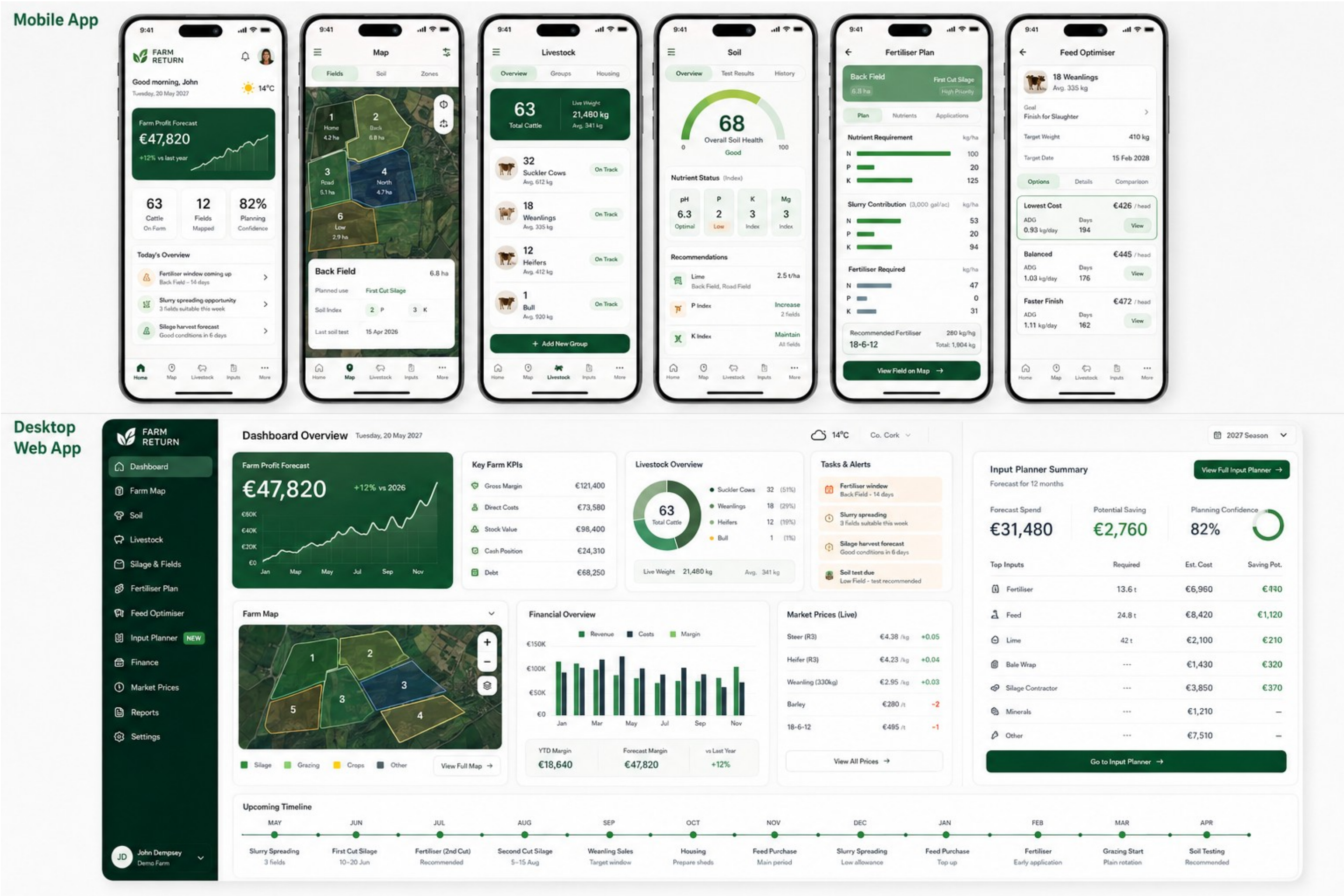
The build is deliberately staged so visual quality, data architecture and scientific accuracy can each be validated before the next layer is added.

Phase	Scope	Exit gate
0 - Repository/design contract	References, docs, route map, tokens, CLAUDE.md, mock farm dataset.	Architecture approved; no feature code yet.
1 - Pixel-accurate UI prototype	All major mobile/desktop screens with mock data and complete navigation.	Visual regression accepted against reference pack.
2 - Central farm data model	Farm, fields, land use, livestock, housing, inputs, finance entities with mock persistence.	Enter-once dependencies proven end-to-end.
3 - Soil/nutrient MVP	Irish soil overlay, editable P/K assumptions, tests, slurry offset, product/cost calculation.	Known test cases independently validated.
4 - Silage/livestock/finance	Silage plans, forage inventory, feed cost estimation, livestock economics, free financial dashboard.	Whole-farm forecast updates when field/livestock data changes.

Phase	Scope	Exit gate
5 - Weather/spreading	Field score, hard stops, forecast windows and application recording.	Rules/data-source review + field-level end-to-end tests.
6 - Input Planner/bulk buying	Forecast demand, stock deduction, demand confirmation, buying-group workflow and saving ledger.	Pilot procurement event can be run from forecast to accepted offer.
7 - Advanced optimiser	Least-cost/profit feed optimiser, market sensitivity, sell/finish scenarios.	Nutrition model reviewed and validated against known scenarios.
8 - Future premium intelligence	Advanced breeding, forecasting, benchmarking, integrations and adviser tools.	Only after free core has strong adoption and validated demand.

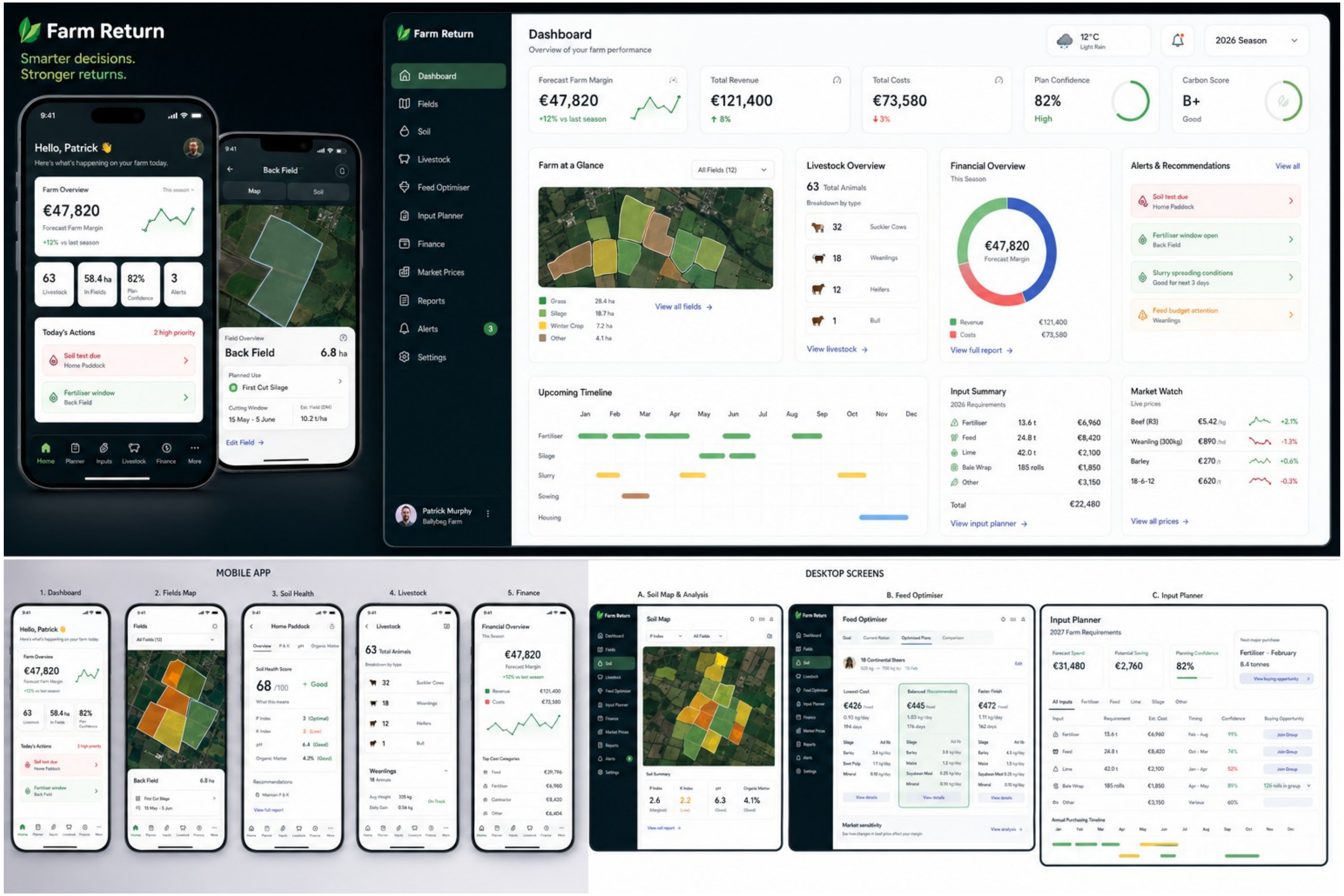
17 Approved visual reference pack

These pages should be copied into /design/reference. They are design references, not sources for production numerical calculations.

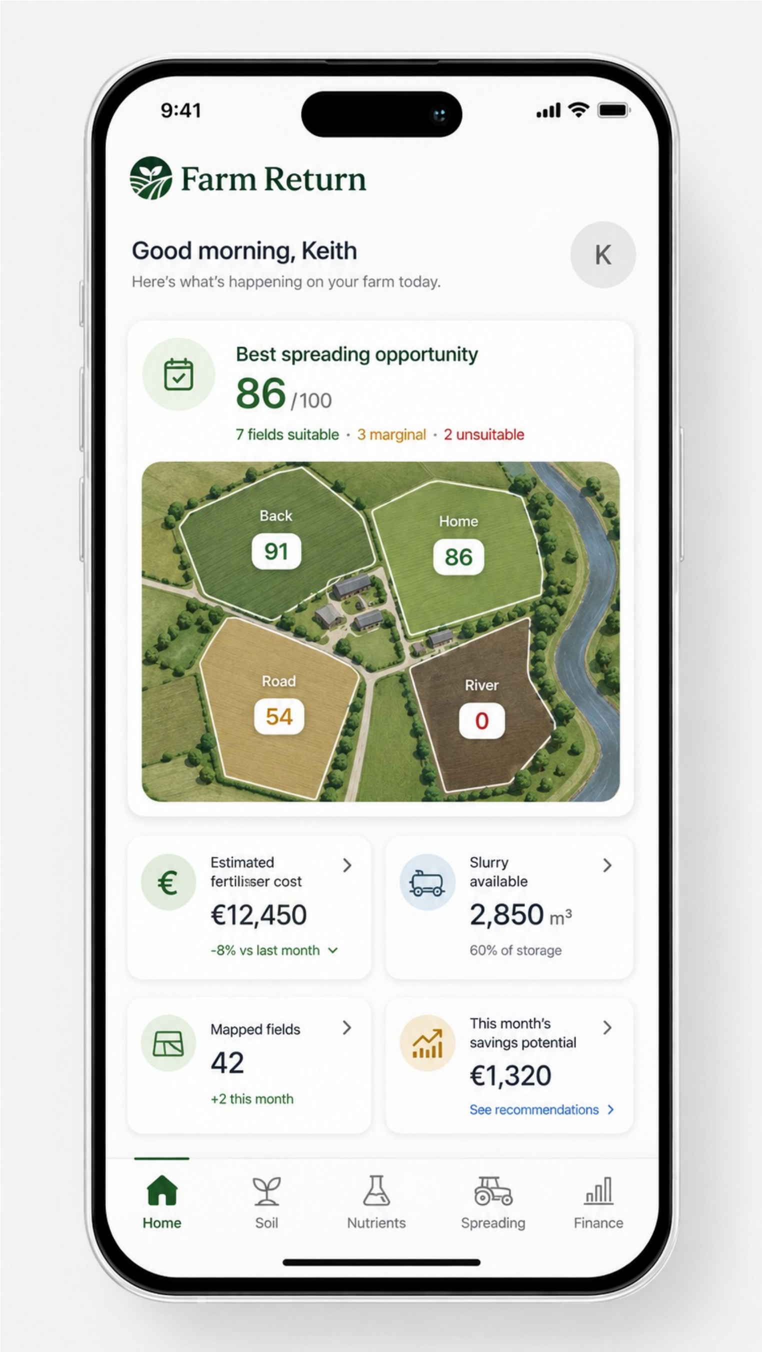


Master reference 1 - complete mobile and desktop Farm Return ecosystem.

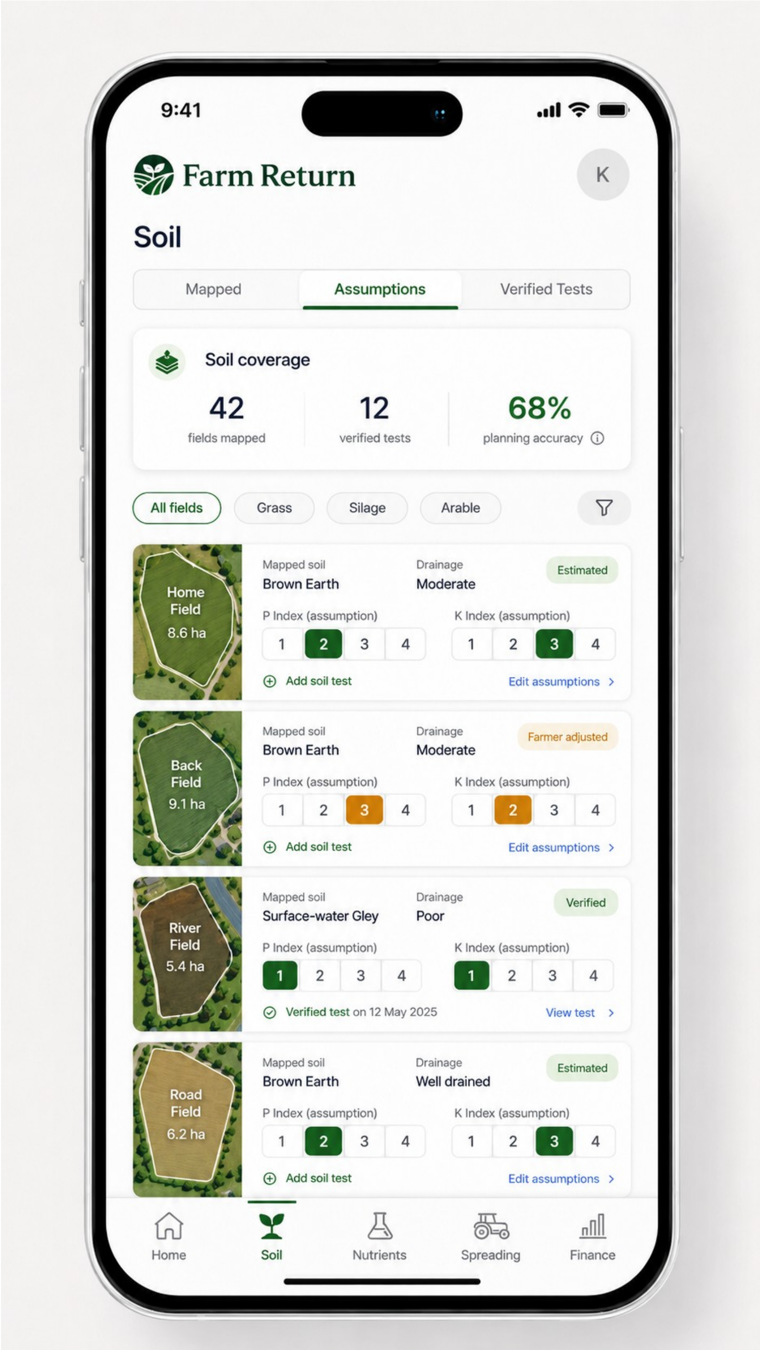
17A Master responsive visual system



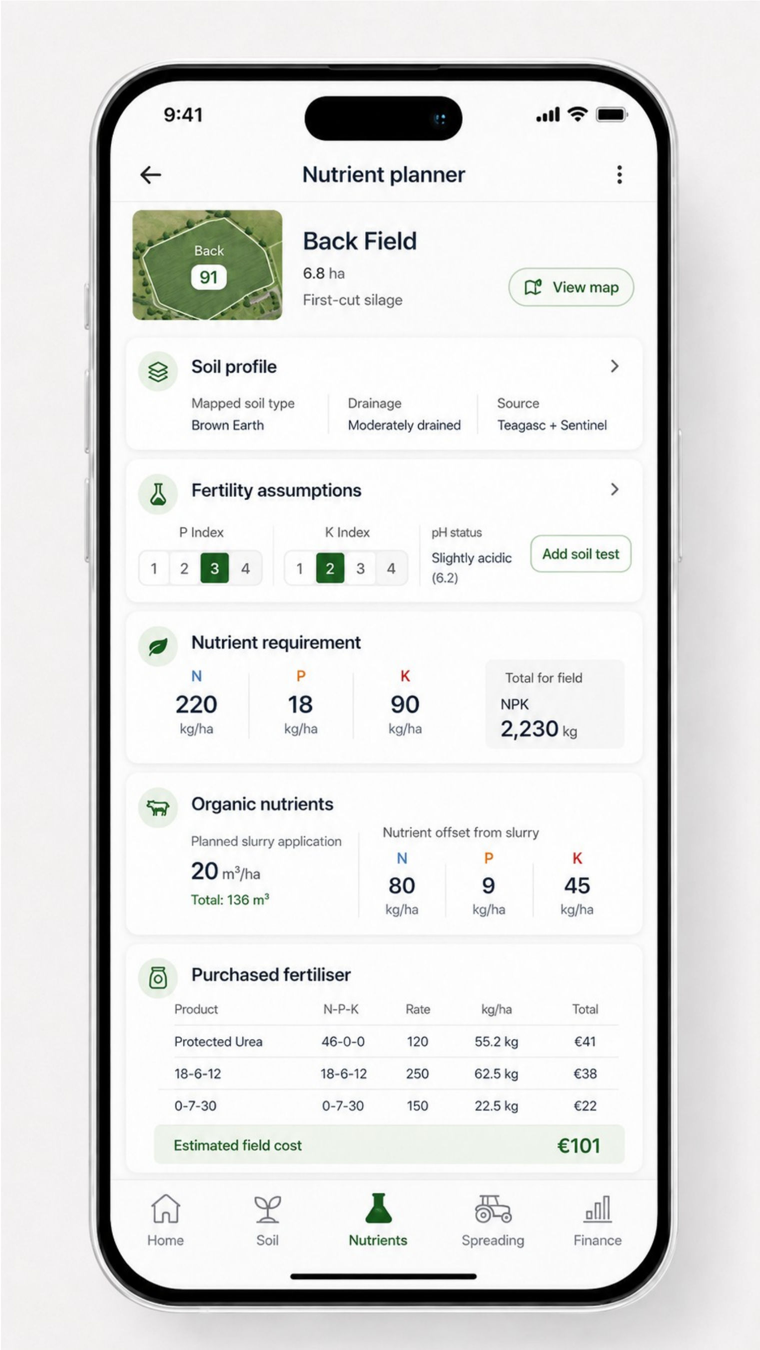
Master reference 2 - high-detail brand and responsive layout system.



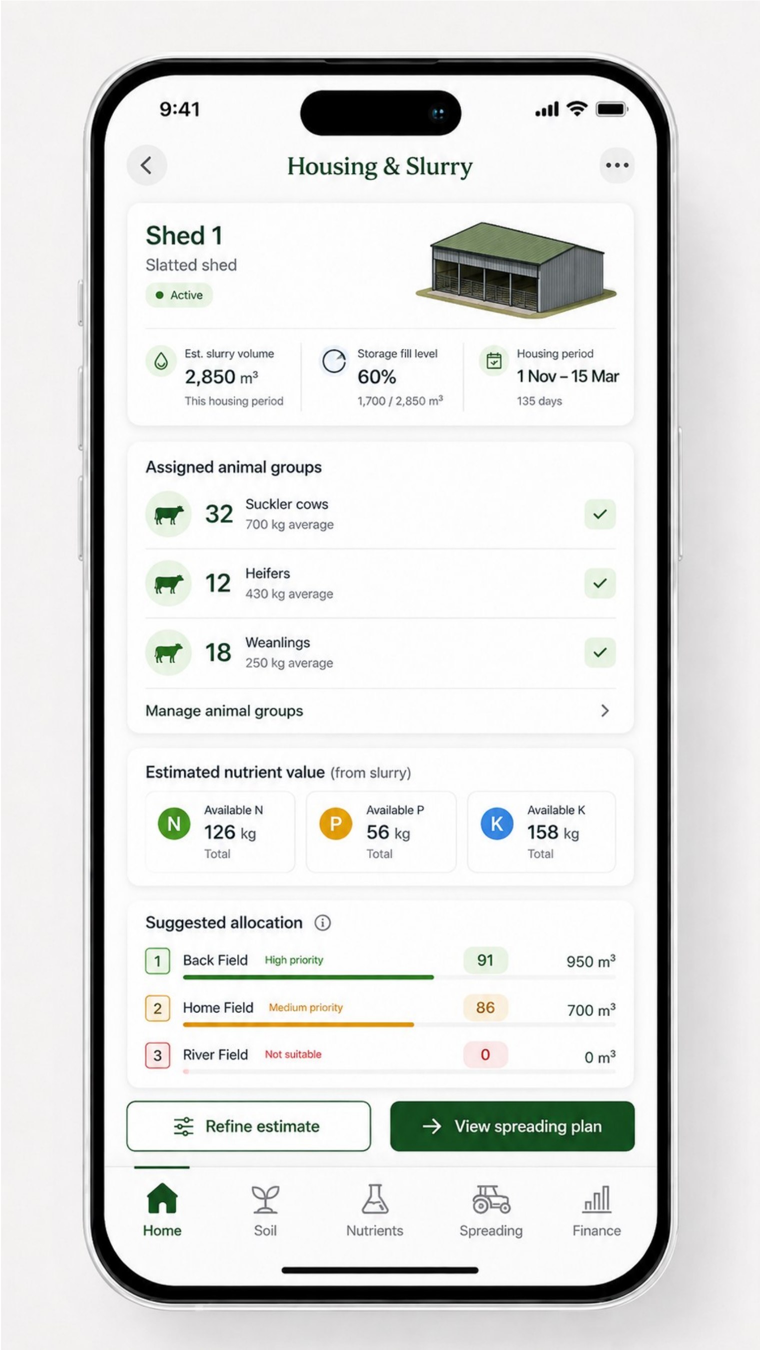
Mobile Home / Dashboard - spreading hero, map, fertiliser cost, slurry, fields and savings.



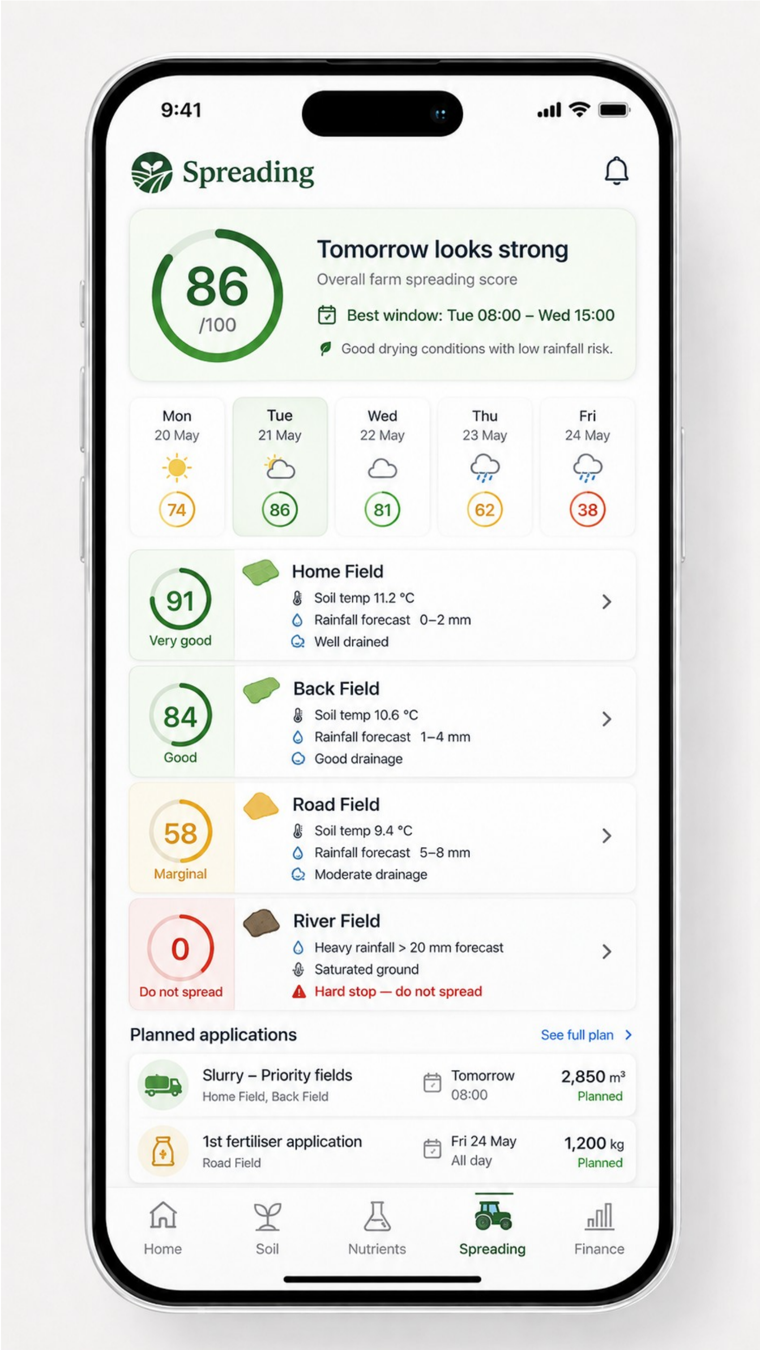
Soil overview - mapped/assumption/verified states and fast P/K adjustment.



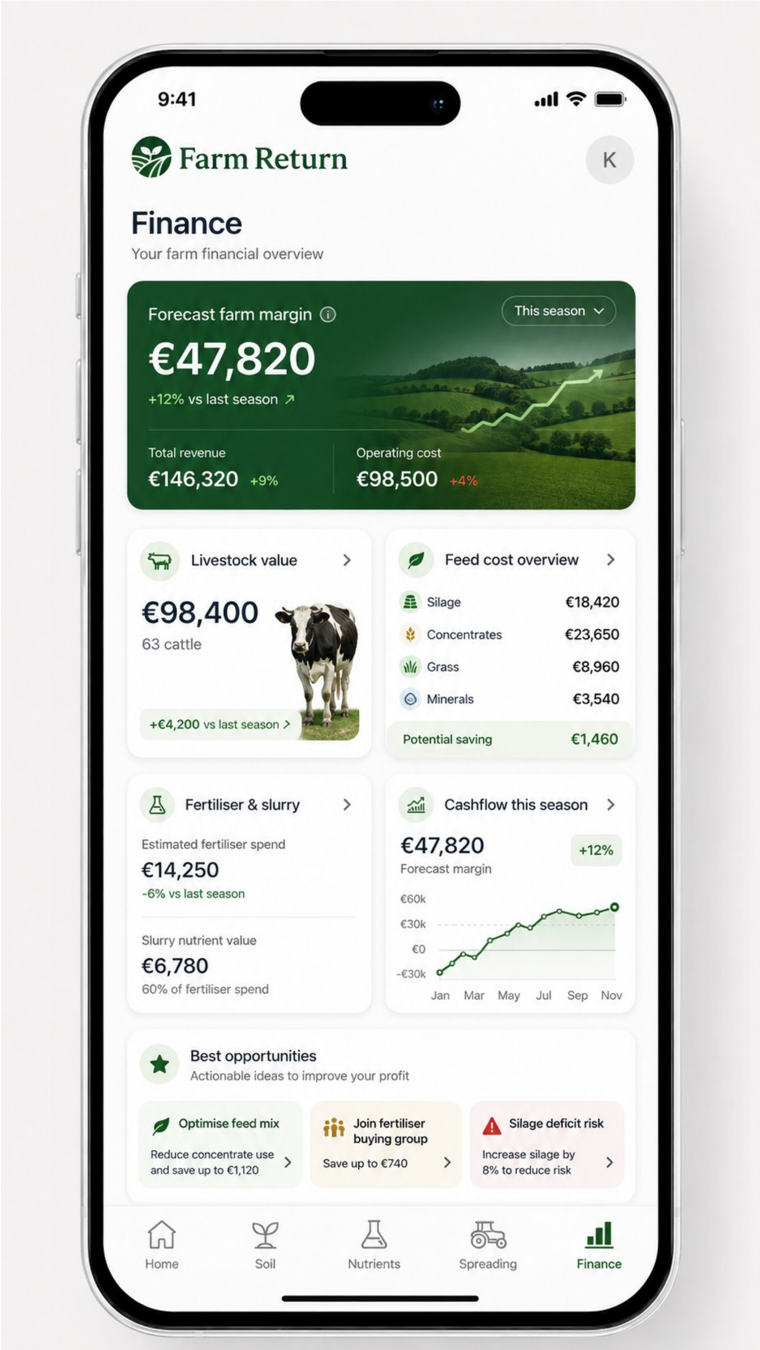
Nutrient Planner - field requirement, slurry contribution and purchased fertiliser.



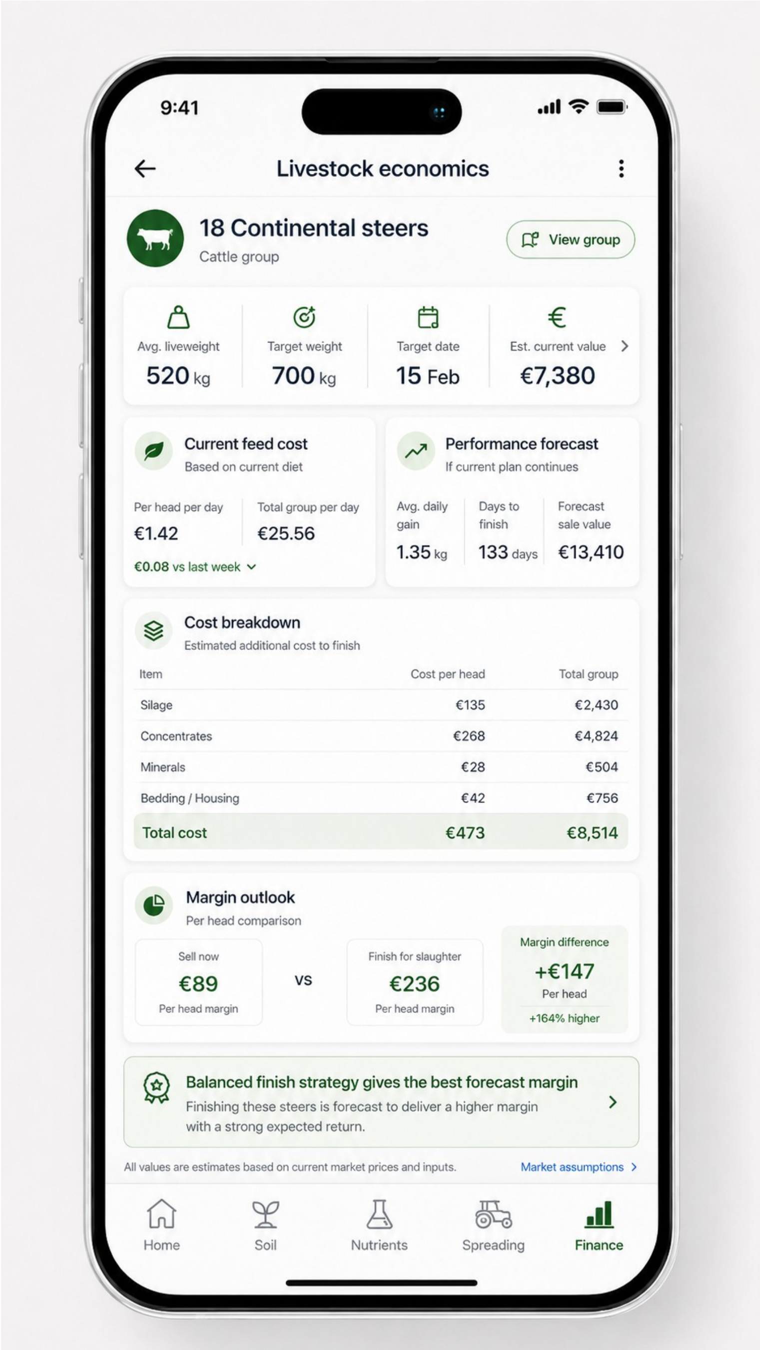
Housing & Slurry - linked groups, storage, nutrient value and field allocation.



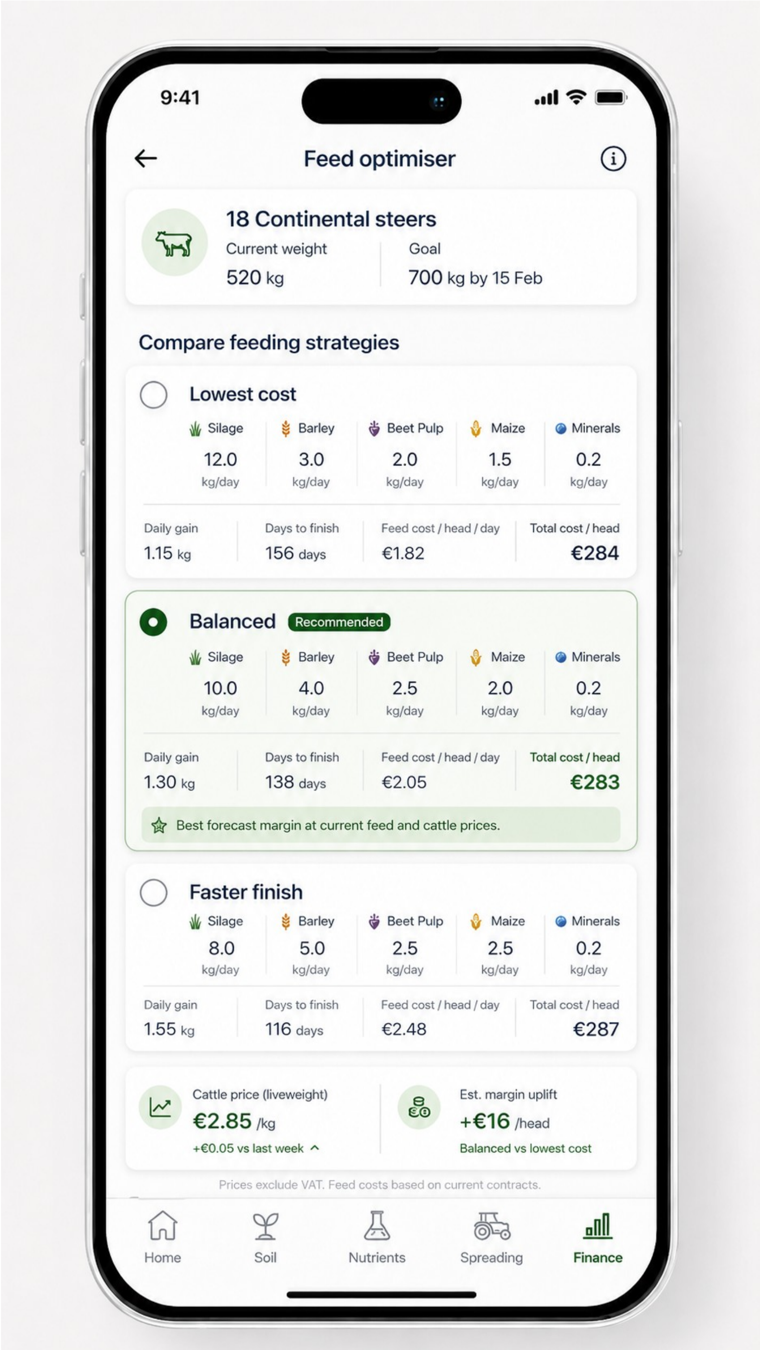
Spreading - field scores, 5-day conditions, hard stop and applications.



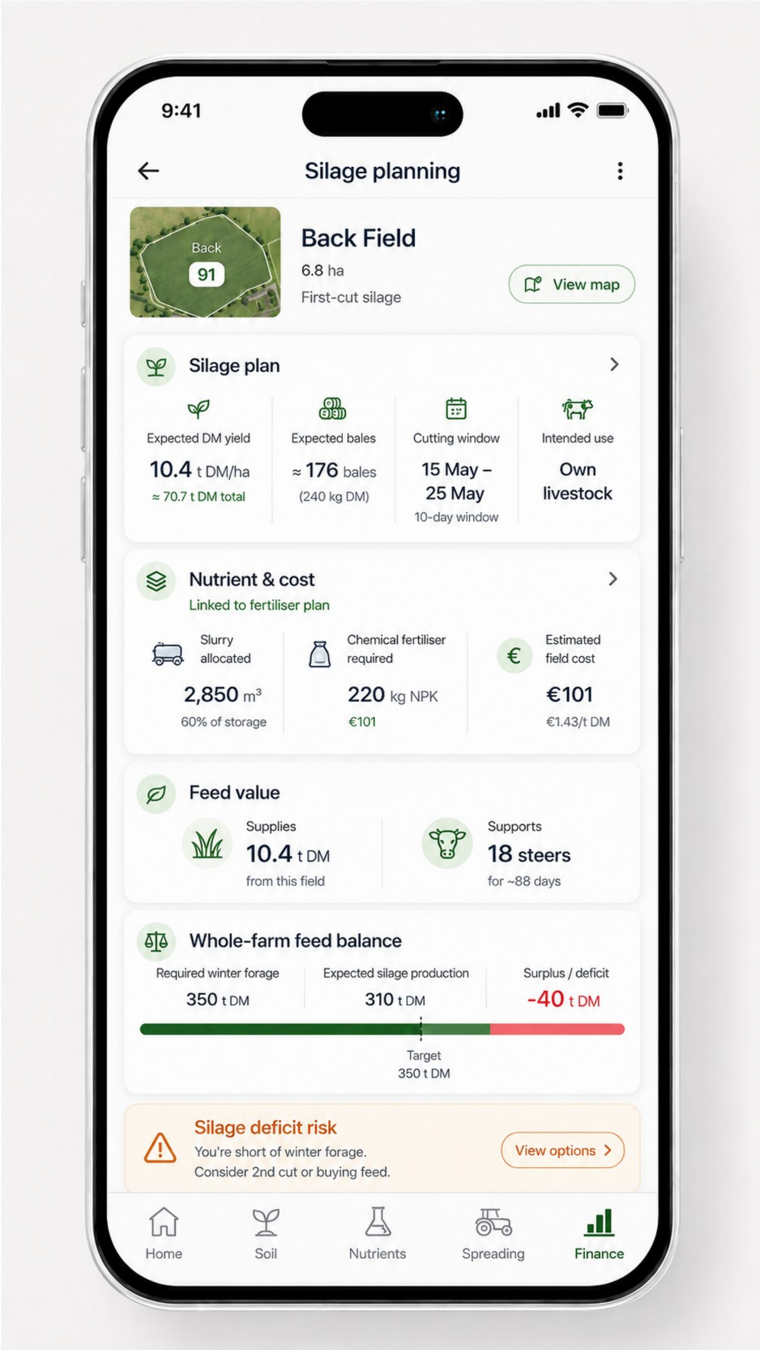
Finance - whole-farm margin, costs, feed, livestock, fertiliser/slurry and opportunities.



Livestock Economics - performance forecast, cost to finish and margin comparison.



Feed Optimiser - least cost, balanced and faster-finish options.



Silage Planning - yield/cut, nutrient cost, feed value and winter forage balance.

18 Authoritative Irish evidence framework

Production rules must be rechecked against the latest published sources at implementation and periodically thereafter. The references below reflect the current framework reviewed for this specification on 23 August 2026.

Teagasc - Fertiliser Advice / Green Book

Core Irish nutrient advice; Green Book 5th Edition remains the baseline, with later crop/enterprise factsheets and current-year guidance layered on top.
https://teagasc.ie/environment/soil/soil-fertility/fertiliser-advice/

Teagasc - Soils, Nutrients and Fertiliser Factsheets (2026)

Current technical factsheets including liming, organic manure, protected urea, first-cut silage and P/K management.
https://teagasc.ie/publications/soils-nutrients-and-fertiliser-factsheets/

Teagasc - First-cut silage planning for quality in 2026

Current example for silage timing and nutrient requirement; includes 100-20-125 kg N-P-K/ha for a 5 t DM/ha example crop and slurry recycling guidance.
https://teagasc.ie/news--events/daily/first-cut-silage-planning-for-quality-in-2026/

Teagasc - Making quality silage on beef farms (2026)
Current first/second-cut nutrient tables and soil fertility/quality context.
https://teagasc.ie/insights/making-quality-silage-on-beef-farms/

Teagasc - Nutritional management of finishing beef cattle (2026)
Current finishing performance, feed efficiency, example beef budgets and cost assumptions; key evidence base for later optimiser validation.
https://teagasc.ie/insights/nutritional-management-of-finishing-beef-cattle/

Teagasc - Irish Soil Information System
National predictive soil map at 1:250,000 with 58 associations and 213 series; physical soil context only, not field fertility verification.
https://teagasc.ie/environment/soil/irish-soil-types-and-maps/irish-soil-information-system/

EPA - Irish Soils Information System research
Background and national soil mapping methodology.
https://www.epa.ie/publications/research/reports/summary-of-findings-epa-research-130-irish-soils-information-system.php

Met Éireann - Farming / Agri-Meteorological Data
Irish rainfall, soil temperature and soil-moisture-deficit information; SMD model distinguishes well, moderately and poorly drained soils.
https://www.met.ie/forecasts/farming

Met Éireann - Agri-Meteorological Data
Technical context for the soil moisture deficit model and drainage classes.
https://www.met.ie/climate/services/agri-meteorological-data

Irish Statute Book - S.I. No. 588/2025
European Union (Good Agricultural Practice for Protection of Waters) Regulations 2025, effective 1 January 2026; current statutory baseline for nutrient/storage/spreading compliance logic.
https://www.irishstatutebook.ie/eli/2025/si/588/made/en/pdf

CSO - Agricultural Price Indices June 2026
Public benchmark datasets include monthly feed-stuff and fertiliser prices and cattle prices.
https://www.cso.ie/en/releasesandpublications/ep/p-api/agriculturalpriceindicesjune2026/data/

Bord Bia - Cattle Trade & Prices
Weekly cattle market data/quotes and price dashboards for market-value context.
https://www.bordbia.ie/farmers-growers/prices-markets/cattle-trade-prices/

Implementation warning
Public scientific and regulatory guidance changes. The application should not encode guidance as permanent constants. Keep rule sets versioned, sourced and updateable, and review licensing/API terms before using external datasets commercially.

Build objective

The implementation standard

A farmer should be able to map a field, assign its purpose, enter livestock and housing once, and then watch Farm Return automatically connect soil, nutrients, slurry, silage, feed, financial forecasts, spreading conditions and purchase requirements. The experience should feel visually premium and operationally simple even though the model underneath is sophisticated. Next action: place this PDF/DOCX and the screenshots into the repository, run Claude in plan mode, and approve the architecture before allowing implementation.